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Deepfakes And Political Propaganda: A Multimodal Study Of Deepfake Imagery And Accompanying Text

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Abstract

The accessibility of free deepfake generation tools poses a growing threat to democratic discourse, as illustrated by the cancellation of the 2024 Romanian presidential elections, driven partly by AI-generated disinformation, underscoring the need for accessible detection tools. Consumer-grade LLMs are a promising option, given their accessibility and the lack of technical expertise required to use them. This study investigates whether such models can be trusted for this purpose by evaluating 3 LLMs (GPT-4o-mini, Gemini-2.5-flash-lite, Qwen-3.5-9B) across six prompting strategies and two languages, English and Romanian, a low-resource language where performance degradation is more likely and understudied. The experiment is conducted over a dataset of 1,200 Romanian political Social Media images and captions, collected and manually annotated for authenticity (real/fake) by the author. Results reveal a strong authenticity bias across all models, as models default to predicting "Real" and often fall below random chance in detecting actual fakes. Among the six prompting strategies explored, including zero-shot, chain-of-thought, persona, and few-shot prompting, the few-shot approach is the only one to consistently overcome this bias, with all six model-language combinations reaching moderate to substantial agreement with true labels, $\kappa = 0.56 - 0.7$. Qwen 3.5 Vision under a simple Romanian prompt, without accompanying social media caption or other outputs requested, provides the best performance, with a sensitivity of 79%, while GPT-4o-mini under the same conditions produces the worst results, with a sensitivity of only 13% (82% under few-shot conditions). This indicates a direct practical concern given GPT's widespread use in Romania, and proves that cross-lingual performance degradation is model-dependent and non-trivial. This study also identifies Romanian diacritic errors as a domain-specific visual indicator of AI-generated content, invoked at nearly twice the rate in Romanian-prompted runs. Finally, this research introduces a set of empirically grounded recommendations for detecting AI-generated content and political propaganda.

Contents

List of Abbreviations	iii
1 Introduction	1
1.1 Motivation	1
1.2 Research Questions	2
1.3 Contributions	3
2 Related Work	3
2.1 Deepfake Generation, Evolution and Detection	4
2.2 LLMs as Deepfake Detectors: Political Content and Misinformation	6
2.3 Human Perception and Cross-Lingual Deepfake Detection in Low- Resource Settings	8
3 Dataset and Experimental Setup	9
3.1 Dataset Collection and Annotation	11
3.2 Inter-Annotator Agreement	14
3.3 Dataset Release	15
3.4 Models	16
3.5 Prompt Strategies (S1-S6)	16
3.6 Evaluation Methodology	20
3.7 Computational Costs & Sustainability	21
4 Results	22
4.1 Deepfake Detection Performance	22
4.2 Sensitivity, Specificity and Cross-Lingual Performance	24
4.3 Propaganda, Manipulation Detection and Political Preference	26
4.4 Visual Cue Analysis	28
4.5 Image-Level Analysis	30
5 Discussion	31
5.1 Effect of Prompt Strategy	31
5.2 Cross-Model Comparison	33
5.3 Cross-Lingual Performance	34
5.4 Romanian-Specific Findings	35
5.5 Model Refusals and Format Compliance	37

5.6	Qualitative Case Studies	37
6	Conclusion and Future Work	39
6.1	Summary of Findings	39
6.2	Limitations	41
6.3	Future Work	42
6.4	Practical Recommendations	43
6.4.1	Visual Indicators of AI-generated/AI-altered Content . .	43
6.4.2	Common indicators of political manipulation	44
	AI Usage Disclosure	i
	References	ii
A	Dataset Examples	vii
B	Images provided as guidance in the Few-Shot prompt	viii
C	Final Prompt Strategies	ix
C.1	S1 – Simple	ix
C.2	S2 – Simple with Context	ix
C.3	S3 – Chain-of-Thought (moderation-safe variant, V4)	x
C.4	S4 – Persona	xi
C.5	S5 – CoT with Persona	xii
C.6	S6 – Few-shot	xiii
D	Additional results	xv
E	Additional visualisations	xvii

List of Abbreviations

Term	Note
LLM	Large Language Model
VLM	Vision-Language Model
CoT	Chain-of-Thought prompting
EN	English prompt condition
RO	Romanian prompt condition
S1–S6	Prompt strategies 1 through 6
(GPT,S1,RO)	Results of the GPT model under S1 conditions in the Romanian prompt
S1	Simple prompt, replicating a query a non-expert user would typically submit (see C.1)
S2	Simple prompt with Social Media caption context added (see C.2)
S3	Chain-of-Thought prompt (see C.3)
S4	Persona-building prompt (see C.4)
S5	Combination of S3 and S4 (see C.5)
S6	Few-shot prompt with 4 labelled example images (2 real, 2 AI-generated) (see C.6)
GAN	Generative Adversarial Networks
Deepfake	AI-generated or AI-altered visual content misrepresenting real people or events
TP	True Positive — predicted fake, actually fake
FP	False Positive — predicted fake, actually real
TN	True Negative — predicted real, actually real
FN	False Negative — predicted real, actually fake
F1	Harmonic mean of Precision and Recall
TPR	True Positive Rate (Recall on fake images)
TNR	True Negative Rate (Recall on real images)
Low-resource languages	languages appearing less in the training corpora of LLMs
GPT	GPT-4o-mini, model used during the experiment
Gemini	Gemini-2.5-flash-lite, model used during the experiment
Qwen	Qwen-3.5-9B, model used during the experiment

1 Introduction

1.1 Motivation

The primary motivation behind this BSc Project emerges from the author's observations of the Romanian social media landscape during and following the 2024 presidential election campaign. The electoral campaign exposed the population to an unprecedented volume of misinformation, a phenomenon not new to Romanian politics, but one significantly amplified by the rapid advancement of large language models and generative AI. The ability to create, alter, or fabricate convincing digital and textual content at scale, combined with the industrialised use of bot networks, has lowered the barrier to political manipulation to a degree previously unseen.

This issue is particularly worrying in Romania, given the country's structural struggles. According to a national study conducted in 2022, over 42% of students aged 7 to 14 are functionally illiterate [5]. At the same time, Romania ranks first across European Union member states in early school dropout rates, at approximately 17% [9] according to data as new as 2025. The low educational levels are directly reflected in reduced media literacy and an increased risk of reading misinformation, making large segments of the population disproportionately exposed to manipulative content.

Unfortunately, the 2024 presidential election exposed these vulnerabilities clearly. Against a high-stakes, deeply polarised geopolitical debate regarding Romania's foreign policy orientation, the first round of elections produced a result that defied all pre-election surveys: a previously unknown candidate, unrepresented in the polls, secured over two million votes, representing approximately 23% of the electorate. The election was later annulled by the Constitutional Court, citing concerns over foreign interference and coordinated digital influence operations [25]. However, the damages are still widely felt within the Romanian society. The society is more divided than ever, whereas the flow of AI-generated and AI-altered political content did not subside following the forced cancellation, as the dataset collected for this thesis illustrates, underscoring the need for robust, scalable, widely available, and trustworthy tools for detecting and flagging artificial imagery in social media contexts, especially in politically sensitive contexts.

The EU AI Act, which imposes regulations and strict rules on how AI-generated content should be marked and used, is still being developed. Regu-

lation is mandatory to slow down manipulation and hate speech, as technology has already outpaced the regulatory timeline.

This work, therefore, investigates whether LLMs can serve as reliable tools for detecting AI-generated and AI-altered political imagery, evaluating their performance across multiple prompting strategies, languages, and model architectures. Beyond benchmark accuracy, the study aims to produce a practical set of visual indicators that users can keep in mind when assessing the authenticity of politically sensitive images in a low-resource language context.

1.2 Research Questions

This project aims to address the following research questions:

1. To what extent can consumer-grade multimodal LLM Models (ChatGPT, Gemini, Qwen) accurately distinguish between authentic and deepfake/AI-generated/AI-altered content?
2. How does detection performance vary when prompts are issued in a low-level language, with less training (Romanian), compared to English?
3. Do structured, and advanced prompting techniques, such as Chain-Of-Thought, Persona-building, and Few-Shot meaningfully improve performance over a simple baseline prompt?
4. How do language-first models (GPT-4o-mini, Gemini-2.5-flash-lite) compare to a vision-first model (Qwen 3.5) across 6 different prompting strategies and languages?
5. Which visual cues do LLMs most frequently invoke when classifying an image as fake, and does explicitly guiding the model’s reasoning affect which cues are signaled?
6. Based on observed model strengths and failure modes, can empirically grounded prompt templates and practical visual indicators be derived to support non-expert users in identifying AI-generated/AI-altered digital content and political propaganda?

1.3 Contributions

The main contributions of this work are:

- A self-annotated dataset comprising 1200 images from Romanian Social Media, labelled for authenticity (AI-altered/Real), alongside a reproducible list of post URLs and ground truth labels available at <https://zenodo.org/records/19686696>.
- An evaluation framework consisting of six different prompting strategies applied across three state-of-the-art LLMs and two languages, a high-resource language (English) and a low-resource language (Romanian), producing a total of 36 experiments on a 1200-image dataset.
- Empirical evidence of cross-lingual performance variation in consumer-grade LLM-based deepfake detection, demonstrating that performance degradation is model-dependent and can be significantly reduced through few-shot prompting.
- Identification of Romanian diacritic errors as domain-specific visual indicators of AI-Generated content, directly measurable across prompting strategies.
- A set of empirically grounded visual cues, derived from observed model behaviour and the author’s personal experience, intended to support non-expert users in identifying AI-Generated content, as well as a descriptive analysis of common propaganda techniques and manipulation cues most frequently identified by the models across the dataset.

2 Related Work

Deepfake detection sits at the intersection of Computer Vision, Natural Language Processing, and media forensics. Given the rapid evolution in AI capabilities during recent years, the barrier to creating convincing manipulated digital content has significantly dropped, making the development of free, reliable, easily accessible detection tools an urgent research problem. While dedicated deepfake detection tools exist for commercial and industrial use, they are typically subscription-based and not designed for occasional, non-expert use, making them inaccessible to the general public who may only need to verify an

isolated image or piece of content. This further motivates the investigation of LLMs as a freely available, general-purpose alternative for detection tasks.

2.1 Deepfake Generation, Evolution and Detection

The term 'Deepfake', derived from the combination of 'deep learning' and 'fake', broadly refers to AI-generated or AI-altered content, consisting of visual or audio content, that misrepresents reality, usually by fabricating, replacing, or manipulating the appearance of real people or events convincingly or deceptively. Manipulation techniques are commonly divided into three categories: facial replacement, facial reenactment and facial synthesis [43]. Each of those exploits different generative mechanisms and presents distinct detection challenges. The underlying technology behind deepfake generation has evolved dramatically over the past decade. Early systems relied on Generative Adversarial Networks (GANs), introduced by Goodfellow et al. in 2014 [15], in which a generator and a discriminator network compete, with the goal of producing increasingly realistic outputs. Early GAN variants suffered from training instability and low output resolution, but the StyleGAN family [22] facilitated synthesis of high-resolution, photorealistic faces, marking a big improvement in deepfakes' realism. The subsequent rise of Diffusion Models, embodied by Stable Diffusion [38] and DALL-E [35], made content generation even more accessible, enabling convincing synthetic imagery from a short text prompt with little to no technical expertise required. Consumer face-swapping tools such as DeepFace-Lab [33] extended this threat to political disinformation campaigns. As Liu et al. [27] describe in their comprehensive survey of the field, the trajectory is best understood as an *"ongoing cat-and-mouse game between generation and detection"*. Last but not least, the boundary between language model and image generator has begun to dissolve: widely available consumer-grade multimodal models such as GPT-4o and Google's Nano Banana are now capable of generating and editing photorealistic images through natural language instructions alone, without requiring specialised tools or any technical knowledge. Nano Banana, in particular, has drawn public attention for its ability to maintain facial consistency across edits, raising specific concerns about the ease with which real photographs of political figures can be convincingly altered.

Unfortunately, the evolution in technology also enabled diversification of its application. As Singh et al. [1] clearly represented in their work, deepfake usage has expanded across multiple domains, including political manipulation,

fraud, non-consensual imagery, and more:

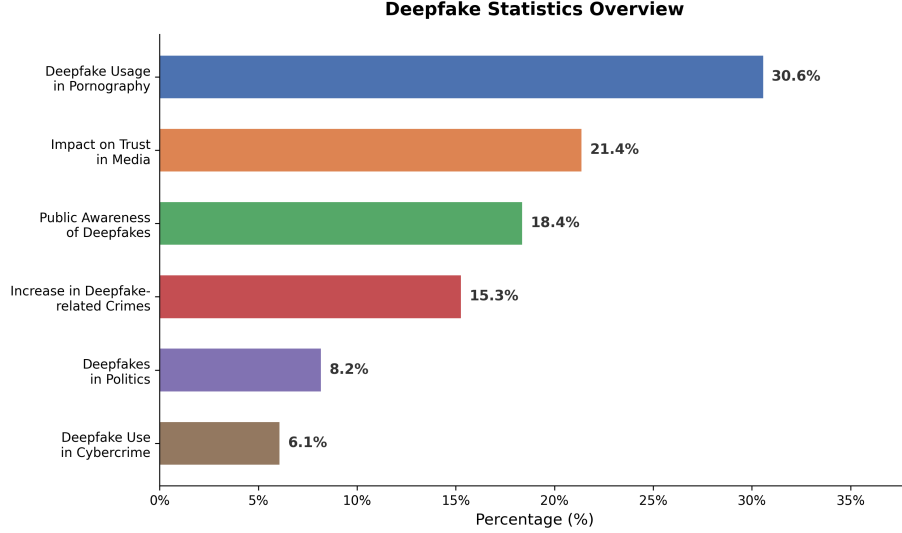


Figure 1: Deepfake Statistics Overview. Data extracted from [1].

Deepfake detection approaches have developed across three primary domains: physical/psysiological (e.g. unnatural blinking, skin colour inconsistencies, etc.), signal-level (detecting high-frequency artefacts left by GAN behind, as demonstrated by F3-Net [27]), and data-driven DNN methods (state-of-the-art Xception-based classifiers or Vision Transformer Models). Despite the evolution, Liu et al. [27] note that generalisation still remains a critical problem, as performance drops sharply on cross-dataset splits and GAN-targeted detectors provide limited coverage against diffusion-generated content.

Current research faces numerous critical limitations. Existing detectors typically analyse pixels in isolation, ignoring linguistic and social cues, such as content captions, identity, and scope of an author. At the same time, standard models are trained on curated benchmark datasets, failing to account for the noise of real-world social media, be it text overlays, varied image quality, or simply the creativity of people driving propaganda. This study addresses these gaps by evaluating consumer-grade LLMs as deepfake detectors across six prompting strategies, from simple zero-shot queries to chain-of-thought reasoning, persona assignment, and few-shot exemplars, on real Romanian political social media content. This approach inherently incorporates linguistic and contextual signals alongside visual analysis, tests whether structured prompting meaningfully

improves detection, and operates under the noisy, compressed, and culturally specific conditions absent from standard benchmarks.

2.2 LLMs as Deepfake Detectors: Political Content and Misinformation

The application of large language models to deepfake detection is a relatively recent but rapidly growing line of research. Jia et al. [18] were among the first to systematically evaluate multimodal LLMs for this task, testing GPT-4V and Gemini 1.0 Pro on face image classification. Their main finding was that context-rich prompts reduced model refusal rates, while more detailed prompts did not reliably improve detection accuracy and increased cost. Notably, most classification errors occurred on real images, suggesting a systematic bias toward labelling content as fake. Their experiment, running simple queries on 1000 real images from the FFHQ dataset [23] and 2000 images created with generative AI models from the benchmark DF³ dataset [20], achieved 50% detection accuracy on real pictures, and above 90% on AI-generated images. ForgerySleuth [42] leveraged multimodal LLMs to detect tampered image regions, finding that GPT-4o achieves strong overall performance in identifying localised alterations. Recent work has extended this evaluation to a broader range of models and tasks. Ren et al. [37] evaluated several LLM’s on a niche deepfake detection benchmark, concluding that *”only OpenAI models have reasonably good performance on our niche deepfake detection task while all other models struggle to have any real distinguishing power”*. The same study probed for hallucination by querying models repeatedly on the same image, raising questions about verdict consistency. In a 2025 study by Tariq et al. [44], a parallel zero-shot evaluation across 4 different models from different providers on a curated benchmark spanning faceswap, reenactment, and synthetic generation provides a useful performance reference for two of the models evaluated here. GPT-4o had a 67% accuracy against three benchmark deepfake datasets, whereas Gemini-2.5-flash went as low as 27%. The work by Pirogov [34], testing VLMs as Zero-Shot Deepfake detectors against benchmark datasets, identifies prompt engineering, including Chain-of-Thought reasoning and flexible prompt design, as a priority for future work, a gap that the six prompting strategies of the present study directly address.

On the political content and dataset side, OPENFAKE [28] introduced a politically grounded deepfake dataset with accompanying social media cap-

tions, extracted from X, Reddit, and Bluesky, focusing on content related to the Canadian Elections in 2025. This dataset is yet to be studied in connection with LLMs’ abilities to correctly identify its contents as AI-generated. The International AI Safety Report from last year [3] concluded that adding warning labels to AI-generated content increased human detection rates only from 10.7% to 21.6%, a modest improvement that raises serious doubts about whether disclosure-based interventions are sufficient. The same study reviews watermarking as a technical countermeasure and notes significant limitations in robustness under real-world distribution conditions. These findings are directly relevant to ongoing policy debates: the EU AI Act is expected to mandate disclosure and watermarking of AI-generated content, yet empirical evidence suggests these measures provide limited protection in practice, motivating the need for automated detection tools that make life easier for the non-expert consumer of AI-generated material.

Beyond model and data selection, the choice of prompting strategy has been shown to significantly impact LLM performance across a range of reasoning and classification tasks. Wei et al. [46] demonstrated that Chain-of-Thought prompting, guiding the model to reason step-by-step before providing an answer, substantially improves performance on complex tasks compared to direct answer prompts. Brown et al. [6] established Few-shot prompting as a great method to improve LLMs’ performance, as it allows models to generalise from a small number of labelled examples without fine-tuning, making it particularly attractive in low-resource settings, such as the Romanian language across this work. Persona-based prompting, in which the model is assigned a domain expert role, has been shown to improve output quality on specialised tasks [41], though there is no consensus among researchers, as Kim et al. [24] refers to this strategy as a ”*double-edged sword*”, where assigning an expert role can introduce overconfidence or constrain the model’s reasoning, ultimately hurting performance in certain classification settings.

Across all those studies, two gaps remain heavily unaddressed. First, all existing evaluation issues prompts exclusively in English, with most studies relying on benchmark datasets that discard contextual signals entirely. According to my literature review, no prior work has evaluated LLM-based deepfake detection in any language other than English, a limitation that becomes particularly significant for politically sensitive material targeting non-English-speaking audiences. Second, while existing work benchmarks model performance on misinformation and manipulated content, it rarely translates empirical findings into actionable

guidance for non-expert users. This study addresses both gaps: by evaluating detection across English and Romanian, a low-resource language absent from prior deepfake research, and by deriving a set of practical visual indicators and prompting recommendations grounded in observed model behaviour across six different prompting strategies.

2.3 Human Perception and Cross-Lingual Deepfake Detection in Low-Resource Settings

Despite significant advances in automated deepfake detection systems, humans remain surprisingly poor at identifying deepfake content. A large-scale study [7] aggregating results from 56 studies across 80,000 participants found that humans correctly identified deepfake images only 53% of the time, barely better than random choice, while performing somewhat better on authentic images at 68%, yielding an overall accuracy of 58%. As per Hameleers et al.[17], factors that complicate human judgement extend way beyond image quality, credibility assessments being heavily influenced by the plausibility of the depicted claim, and that deepfakes presented in a foreign context, "*US Deepfakes presented to Dutch people*", where audiences lack the background knowledge to assess authenticity, are rated as significantly more credible than those featuring familiar figures or settings. This has direct implications for political disinformation targeting audiences outside the country of origin. The political stakes of this detection gap are considerable. AI-generated content can be produced at an unprecedented scale and at a fraction of the cost of human-generated material, and studies have found it to be at least as persuasive as human-generated political messaging under experimental conditions. The International AI Safety Report [3] from last year quoted a 2024 survey which proved majority of Americans across the political spectrum expressed great concern about AI being used to fabricate information about election candidates. This concern is definitely applicable to citizens of other countries, reflecting the broad, general global anxiety caused by the rapid development of AI. The same report notes that AI-generated manipulative content is perceived as less accurate yet shared at similar rates to authentic content, suggesting that popularity is largely decoupled from perceived credibility. A further dimension still largely understudied in the NLP research is multilinguality, cross-linguality, and performance in low-resource settings. The overwhelming majority of LLM-focused research, including Deepfake detection work, is conducted in English, on English-language datasets, with English prompts. This is

a significant blind spot given that disinformation campaigns are inherently local, as they target specific linguistic communities, exploit culturally specific figures, and circulate on platforms dominated by the local language. Romanian presents a particularly relevant case. The language accounts for approximately 0.6% of Common Crawl [31], the primary pretraining corpus for most LLMs, making it one of the most underrepresented EU languages in deployed LLM systems, and a natural candidate for studying cross-lingual performance degradation. The cross-lingual element is further complicated in this study’s setup, where the accompanying post text is always in Romanian regardless of the prompt language, a deliberate design choice that mirrors real-world conditions rather than sanitising them through translation.

3 Dataset and Experimental Setup

This chapter describes the dataset, models, and prompting strategies used in the experiments. The study evaluates six prompting strategies across three large language models and two languages (English and Romanian), yielding a total of 36 experimental configurations: a simple zero-shot prompt (S1), a simple prompt with social media caption context (S2), a chain-of-thought prompt (S3), a persona-building prompt (S4), a combination of S3 and S4 (S5), and a few-shot prompt with four labelled example images (S6). Figure 2 provides an overview of the full experimental pipeline.

Project & experiment overview

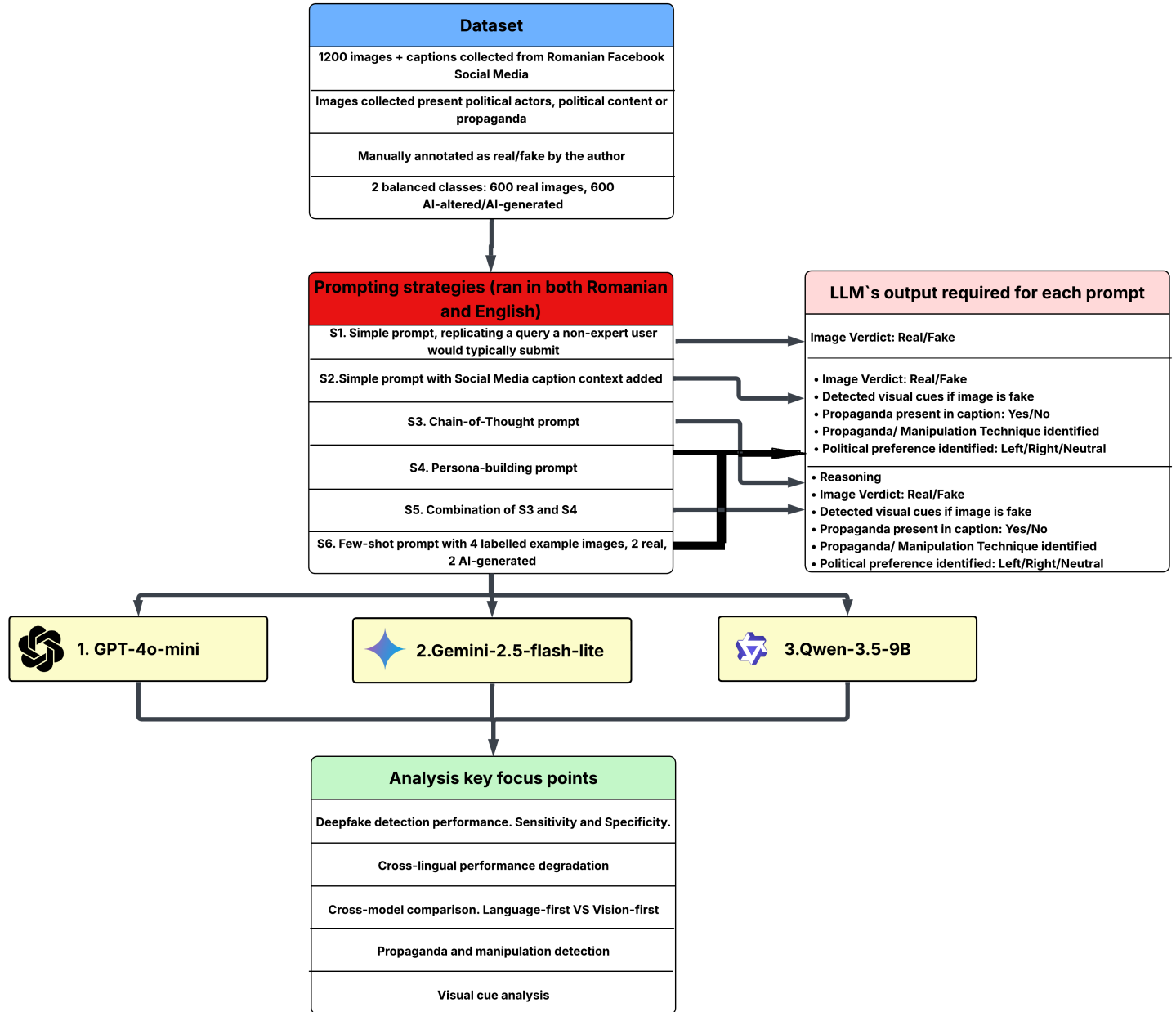


Figure 2: Overview of the experiment, illustrating the flow from dataset construction through six prompting strategies, evaluated across three large language models (GPT-4o-mini, Gemini-2.5-Flash-Lite, Qwen-3.5-9B) in both English and Romanian, yielding 36 experimental configurations.

3.1 Dataset Collection and Annotation

The dataset used for this experiment contains 1200 image-caption pairs, collected from Romanian Social Media (Facebook) during the months of March and April 2026. The data is split evenly into 600 AI-generated/AI-altered images (therefore referred to as 'Fakes') and 600 real, authentic images. All the images collected originate from pages with a political affiliation (e.g. Political Parties, Politicians, Popular influencers discussing political topics, etc.), though the content varies: while the majority of the images depict Romanian Political Figures directly, the dataset also includes AI-generated imagery used for political satire, promotion of political initiatives, as well as social commentary (e.g. caricatures, symbolic images and proverbs). This diversity in the images and the associated context reflects the full range of AI-generated content with a Political message circulating on Romanian political Facebook pages. For each of the 1200 images collected, the accompanying post caption was recorded in Romanian and no filtering or shortening steps have been taken. The caption often includes hashtags, emojis, and other elements of text characteristic of social media text. Not all posts/ images included an accompanying caption, as 150 (accounting to 12.5% of the total dataset) of the visual content had no text associated, distributed across both classes, 65 in the Fake set and 85 in the Real set. While this could have been further limited by an even more extensive search across the feed, this reflects real-world social media conditions and introduces a natural variation in the contextual information available to the models across S2-S6 runs.

Fake images were identified through two complementary methods: organic discovery via Facebook's algorithm's recommendations on the feed, where AI-generated political content surfaced regularly, from both followed pages and unknown pages, and targeted browsing of pages associated with both right-wing and centrist political movements and supporters. A few examples are mentioned below:

Page/Account	Description	Orientation
AUR	Alliance for the Union of Romanians, leading far-right party	Far-right
George Simion	Leader of AUR	Far-right
Călin Georgescu	Winner of the first round of the 2024 presidential elections, independent far-right politician	Far-right
Mihai Tiron	Influencer supporting AUR and critical of current government leadership	Far-right
Vlad Craioveanu	Influencer supporting current leadership, critical of far-right movements	Centrist-left. Progressive
Lucian Mândruță	Journalist and influencer supporting current leadership, critical of far-right movements	Centrist-left. Progressive

Table 1: Example of pages and accounts used as sources for fake image collection.

It is worth mentioning that Romania’s current political landscape lacks a prominent left-wing or far-left party. The majority of the population and political discourse sits within a left-centrist to right range, with the far-right representing, at the time of this research, the primary opposition force. This context explains the absence of clearly left-wing sources in the dataset and further supports the ideological skew observed in the fake set. The images belonging to the fake set span across the last two years of content (2024-2026), reflecting the period in which LLM generation and alteration of AI content became widely available to non-expert users. The real images were all sourced from the Facebook pages of the Romanian President, Prime Minister, leaders of the opposition, other prominent political figures, and various government institutions, with images spanning across 6 years, 2020-2026.

All images were annotated by a single annotator, the author, for authenticity (AI-generated/AI-altered as "fake" vs. real). While prior exposure to AI-generated content informed the annotation process, bias in classification cannot be fully excluded, particularly given the author’s familiarity with the content promoted by the source pages and the implicit political preferences of both the Facebook poster and this study’s author. The primary limitation of this annotation approach is the use of a single annotator. To mitigate this, inter-annotator agreement is reported in Section 3.2.

The fake set exhibits an ideological skew toward right-wing and nationalist (or "Sovereignist") content. This reflects the observed reality of AI-generated

image production on Romanian Social Media, where right-wing and Sovereignist pages predominantly deploy AI-generated imagery for political manipulation, fabricating scenes, altering appearances, or constructing false narratives, whereas left-centre or people with progressive political beliefs more commonly use AI-generated content for satirical purposes, data and statistics visualisation, or presentation of projects. This pattern is consistent across Europe, where far-right movements have been documented as the drivers of the distribution of AI-altered content for voter engagement and political messaging [4]. Therefore, this rather represents an inherent characteristic of the phenomenon examined during this research rather than a deliberate collection bias. Nevertheless, it represents a limitation of the dataset’s representativeness and should be taken into consideration both when replicating this experiment and during the interpretation of the results.

Several decisions had to be made during the collection of the images in order to ensure dataset coherence. Collage examples (where a single image was composed of two or more smaller images), and clearly absurd (e.g., the Romanian President as a dog) or satirical fakes were retained as they represent real circulating content and constitute a meaningful test of the model’s detection capabilities. Images containing real, verifiable text or statistics pasted on top of deepfake images, as well as real images AI-altered with text on top, were also included. Images self-labelled by the posters as satirical (“pamphlet”/“pamflet”) or AI-generated, as well as images containing the Gemini generation watermark or marked by Facebook as ‘Read more about AI content’, were also included. The ground truth for those images is therefore easy to detect, and the models’ ability to leverage visible textual cues present in images will be briefly analysed in Section 5.6. It is worth noting that fewer than 50 images across the dataset fall under any of these categories, reflecting the relative rarity of explicitly labelled AI-generated content in the wild. Finally, 100 real images and 100 AI-generated images from flickr-and-genai-dataset-ff-genai [2] were initially added to establish some baseline metrics, but were later dropped as near-zero model accuracy on these images suggested a significant distribution shift from the social media context the models appear to rely on.



Fake image — scene incoherence and contextual implausibility.

Caption (translated): “Radio Erevan version: In recognition of his dedication to defending democracy and the rule of law, the Prime Minister of Romania was escorted in flight by the Secret Service.” (Radio Erevan is a Soviet-era satirical joke format, implying deliberate irony.)



Fake image — contextual implausibility, digital artefacts and textual errors.

Caption (translated and shortened): “When the President Appropriates Justice. The President of Romania, Nicușor Dan, has decided that his constitutional role is no longer sufficient. [...] The Constitution does not grant him the right to gather the frustrations of judges and prosecutors, place them under a presidential letterhead, and turn them into an official document. This is not reform. This is interference.[...]”



Real image.

Caption (translated): “I have been coming to Alba Iulia every year since 2004. This year we were the most numerous! There were no incidents among the sovereignists — they simply did not let us all be together! The time of the ROMANIANS has come!”

Figure 3: Example image-caption pairs from the dataset, with EN-translated captions.

More examples from the dataset, including clear fakes, satirical images, creator-labelled content, and collages, can be seen in the **Appendix A**.

3.2 Inter-Annotator Agreement

To validate the author’s annotation, two independent annotators were asked to classify a set of 50 distinct images (25 fake and 25 real according to the authors’ annotation), without access to the author’s annotation, the caption of each image, or the poster’s name or political orientation. Neither annotator holds formal accreditation or certifications in deepfake detection, but they are both Romanian, familiar with the Romanian Social Media, therefore reflecting the perspective of an informed general user rather than a domain expert.

The first annotator produced 5 disagreements out of 50 images, yielding a Cohen’s Kappa of 0.8. The second produced 6 disagreements, yielding a Kappa of 0.76. Both scores fall within the ”substantial agreement” range as defined by Landis&Koch [26], indicating that the author’s labels are likely reproducible by independent observers. It must be acknowledged that using only 100 images, 8.3% of the full dataset for inter-annotator validation, is a limitation of this study, due to time constraints. Full independent annotation of the 1200-image dataset remains subject to future work, along with other annotations such as the presence of propaganda/manipulation, manipulation technique, etc.

3.3 Dataset Release

Due to copyright restrictions under Facebook’s Terms of Service [30] [29] and privacy obligations under the General Data Protection Regulation (GDPR[36], Regulation EU 2016/679), the images and captions collected cannot be released as initially planned. Republishing images sourced from Facebook would constitute a violation of the platform’s intellectual property terms, and the presence of recognisable individuals in political content raises additional data protection concerns under Article 17 GDPR [10] (right to erasure). While Article 89(1) [12] provides a research exemption allowing scientific studies (like this one) to override erasure obligations under appropriate safeguards, this exemption applies only to the processing of data, not to its publication. Images and accompanying captions were downloaded manually and stored locally for the duration of this research to ensure consistency of results across all prompting strategies and languages, and to prevent potential data loss due to post-deletions or alterations occurring during the experimental period. In accordance with GDPR data minimisation principles (Article 5 [11]), all images and captions will be deleted upon completion of the study.

In order to ensure reproducibility and allow for a visualisation of the images included in this experiment, the dataset is released as an annotated index containing the original post URL, the URL of the poster’s page and the manually assigned authenticity labels (”Yes” for fake, ”No” for real), available on both Zenodo (at DOI: [10.5281/zenodo.19686696]) [13] and Github [14] under a CC BY 4.0 license. The license applies only to the annotations, the author’s own intellectual contribution, and does not extend in any way to the linked content, which remains subject to Facebook’s Terms of Service [30][29], and the rights of the original content creators in compliance with applicable data protection reg-

ulations, including GDPR [36]. Researchers wishing to replicate this study are responsible for retrieving images and captions directly from the provided URLs in compliance with applicable data protection regulations, including [36], and Facebook’s Terms of Service [30][29]. Some posts may no longer be available at the time of retrieval as content may have been restricted, altered, or deleted by either the users or the platform since the time of collection during this study.

3.4 Models

Three state-of-the-art vision-language models (VLMs) were selected for this study: **GPT-4o-mini** [32] by OpenAI, **Gemini 2.5 Flash Lite** [16] by Google, and **Qwen3.5** [45] by Alibaba. GPT-4o-mini and Gemini-2.5-flash-lite are primarily language-first models with strong multimodal capabilities, while Qwen3.5 is considered a visual-first model, more visually grounded. This model selection choice was made to enable cross-architecture comparison. The mini, lite, and flash variants were chosen with both budget and sustainability in mind. More capable models such as GPT-4o and Gemini 2.5 Pro are significantly more expensive and consume considerably greater computational resources (e.g., GPT-4o-mini is 15-20× cheaper per million of tokens). Testing such models was outside the scope of this study, both for budgetary reasons and to limit the environmental footprint of the 36-condition experimental setup. This is acknowledged as a limitation, as more capable models may yield better results, especially on subtle or ambiguous fakes. Future work employing larger models should consider reporting energy consumption alongside monetary cost, which this study addresses in Section 3.7. Meta’s Llama 4 was also initially considered as the third model. Llama 4 Maverick was not available as a serverless inference option on TogetherAI, the platform selected for third-party model access, while Llama 4 Scout was available at the outset but deprecated by the platform prior to the start of the study. Qwen3.5 was selected as a replacement.

3.5 Prompt Strategies (S1-S6)

Six prompting strategies were designed and evaluated across all three models and both languages, ending up in 36 experimental conditions in total. The strategy design is motivated in part by gaps identified in prior work: [34] explicitly identifies chain-of-thought reasoning and flexible prompting approaches as directions for future research in LLM-based deepfake detection, while [37] demonstrates

that detection performance varies substantially across model families, underscoring the need for a systematic, multi-model evaluation framework. The strategies progress from minimal to increasingly structured and information-rich prompts, allowing for a controlled assessment of how prompt complexity and design affect detection performance across architectures and languages.

All strategies share a common structured output format, with the number of output fields and the allocated **output_max_tokens** varying by strategy, and with complexity increasing across strategies. The simple strategy, **S1**, returns a single binary verdict (Yes/No) indicating whether the image is AI-generated or AI-altered. The simple strategy that also includes the caption **S2**, the persona-building strategy **S4**, and the few-shot prompt **S6** return the following five structured fields:

1. **Image decision:** [Yes/No] (Is the image AI-generated/deepfake/AI-altered?)
2. **Detected visual cues:** [a/b/c/d/e/N] — [all that apply, comma-separated] a. Anatomical implausibility / b. Scene incoherence / c. Contextual implausibility / d. Digital artefacts / e. Textual errors in image / N. Real image (no cues detected)
3. **Political propaganda or manipulation:** [Yes/No]
4. **Manipulation or propaganda technique:** [1/2/3/4/5/N] — [all that apply, comma-separated]: (1) emotional appeal, (2) name-calling or smear, (3) false context or decontextualisation, (4) appeal to authority, (5) simplification, (N) not applicable
5. **Political preference:** [Left/Right/Neutral]

The Chain-of-Thought strategies **S3** and **S5** return the same five fields with the addition of a preceding reasoning step: **Step-by-step reasoning** : a **3–5 sentence free-text** summary of the model’s observations, prior to the structured classification output.

The list of possible detected visual cues, used across S2–S6, **a. Anatomical implausibility, b. Scene incoherence, c. Contextual implausibility, d. Digital artefacts, e. Textual errors in image**; was derived empirically. A set of 15 fake images was presented to four independent non-expert users, who were asked to provide a binary classification judgment with respect to the image’s authenticity and a short free-text reasoning for each image. The

reasoning responses were analysed by the author, yielding nine recurring detection categories: subject-context relation, contextual incoherence, background anomalies, scale and proportion errors, physical and anatomical implausibilities, unnatural colours and compression artefacts, object unrealisticness, unreadable background text, and prior knowledge of the event, person, or context. These were consolidated into the five cues mentioned above (a-e) to reduce output complexity and avoid exceeding model token budgets. The propaganda and manipulation technique taxonomy appearing as model output between S2-S6 follows the classification scheme introduced in SemEval-2024 Task 4 [8], which defines five persuasion techniques commonly employed in political misinformation: **1.emotional appeal** (persuading through emotion rather than fact or logic), **2.name-calling or smear** (discrediting a person or group by attaching a negative label), **3.false context or decontextualisation**, **4.appeal to authority** (lending credibility to a claim by invoking the real or fabricated endorsement of a respected figure or institution), and **5.simplification** (reducing a complex issue to a single cause, omitting nuance).

S1-Simple (see C.1) provides the model with only the image and a simple question to classify it as authentic or AI-generated, returning a binary verdict: "Yes"/"No". This is meant, in a more structured way, to replicate the way most users would query a LLM and serves as a baseline metric against which all other strategies will be measured. Given the minimal output, **output_max_tokens** was set to 10.

S2-Simple with Context (see C.2) extends the simple prompt from S1 by including the accompanying post caption as additional input, which is always in Romanian, irrespective of the language the prompt was written in. Translation of the context was considered, but maintaining the original caption was preferred to not include any potential translation bias and to include a cross-lingual element in this research. S2 returns five structured fields: image decision, detected visual cues, propaganda flag, manipulation technique, and political preference, using a **output_max_tokens** budget of 150.

S3-Chain-of-Thought (see C.3) is built based on the Wei et al. work [46], and it instructs each model to reason step-by-step through a set of visual and contextual cues before producing the structured output. This prompt adds a reasoning field, restricted to 3-5 sentences. The reasoning steps guide the model to attend to anatomical inconsistencies, digital artefacts, lighting coherence, contextual plausibility, and Romanian diacritic anomalies (ă, â, î, ș, ț), which, across the dataset collection, have been observed to be a frequent indicator of

AI-alteration, as AI image generators frequently render them under different fonts, colours, or wrong signs. More in-depth reasoning logic is also included for the Propaganda/Manipulation detection, explaining each manipulation taxonomy in detail. The original S3 English prompt triggered GPT-4o-mini’s content moderation filters at approximately 40% on a preliminary 10-image sample, due to explicit political manipulation language in the reasoning instructions. A moderation-safe variant was produced by softening three CoT steps while preserving the intent of each instruction. No such issue was observed in the Romanian prompt or with Gemini and Qwen, suggesting GPT’s content moderation is more sensitive to explicit political language in English, a finding discussed further in Section 5.5. The S3 prompt had **output_max_tokens** budget of 500, to allow enough output tokens for the reasoning field.

S4-Persona (see C.4) assigns the model the role of a professional digital forensics analyst with experience in the Eastern European political environment, priming it to approach the task with more confidence and a presumed domain expertise. The reasoning field was dropped for this strategy. During preliminary testing, Qwen produced unsolicited analytical notes when prompted in Romanian, including a ”Notă analitică” = Analytical Note that would go beyond the 150 **output_max_tokens** set for this prompt. An explicit prohibition line was added to the Romanian prompt to restrict this behaviour.

S5-CoT&Persona (see C.5) combines S3 and S4, applying both step-by-step reasoning structure and the forensic analyst persona at the same time, resulting in one very complex prompt. The **output_max_tokens** was set to 500, consistent with S3, given the same output format is expected from the LLMs.

S6-Few-shot (see C.6) provides four fixed example image-caption pairs, two fake and two real, before the query image. The examples feature two pictures of the Romanian President and two pictures of one of the most prominent Sovereignist figures in Romania, one real and one fake, each, teaching the model that identity alone is not a reliable detection signal. Their names are not disclosed in the prompt to prevent identity-based shortcuts. The examples were also selected to cover different visual and manipulation cue types, providing a richer and more generalisable signal. Due to the presence of four base64-encoded example images in each of the 1200 prompts, S6 batch files are approximately five times larger than S1–S5, requiring additional splitting during batch submission (15 splits for GPT and Qwen, 3 for Gemini) and consuming proportionally greater computational resources. The **output_max_tokens** budget for S6 was

set to 600. The examples used, their expected model outputs, and the corresponding images are provided in Appendix B.

Several design decisions were made for all strategies. EXIF metadata was stripped from all images before base64 encoding to prevent models from inferring ground truth labels from embedded filename (as all fake images are named "1.jpeg-600.jpeg" locally) or metadata patterns. The instruction "no external knowledge required" was deliberately excluded, as the LLMs are encouraged to use the best of their knowledge when making a decision. The justification field present in early prompt versions was dropped from the final S2 and S4 designs, as it risked producing mechanical restatements of already-selected cues, and the reasoning resulting from S3, S5, and S6 enables enough analysis. Before each full batch run, every prompt variant, in both English and Romanian, was tested on a preliminary sample of 10 images to verify output format compliance, identify parsing failures, and detect content moderation conflicts before committing to the full 1,200-image dataset. The final prompt designs were reached through iterative development across multiple versions per strategy, each addressing output compliance issues or content moderation conflicts observed in preliminary testing. The full prompt evolution history and the final Romanian prompts can be found in the GitHub Repository [14]. The complete final prompt text for each strategy in English is provided in Appendix C.

3.6 Evaluation Methodology

All experimental runs were submitted via the Batch API of each respective provider: OpenAI Batch API for GPT-4o-mini, Google Vertex AI for Gemini 2.5 Flash Lite, and TogetherAI for Qwen3.5-9B. Batch API submissions are 50% cheaper than standard API calls and guarantee responses within 24 hours; in practice, all batches were completed within approximately 1.5 hours. Due to provider-imposed file size limits (100MB for OpenAI and TogetherAI, 500MB for Vertex AI), S1–S5 required 4 splits per language for GPT-4o-mini and Qwen3.5 and a single file for Gemini. S6 required 15 splits per language for GPT-4o-mini and Qwen3.5, and 3 for Gemini, due to the embedded example images. Output parsing had two filters: a primary labelled regex parser, followed by a compact fallback for partially malformed responses observed. Unparseable rows were flagged as UNPARSED and excluded from metric computation. The rejection rates across models and strategies were very low, despite the political nature of the images and captions. The GPT model had the highest rejection rate,

with S5 in English having the highest rejection rate at 13%, 160/1200 prompts. Qwen’s verbose reasoning mode was disabled via `enable_thinking = False`, as the model tended to use more than 2000 output tokens for S1 during the first testing. The temperature was set to 0 across all models and strategies for reproducibility and to capture the LLMs under the standard settings users use. To prevent cross-image context contamination, a new conversation thread was initiated for each image across all prompting strategies and models. Performance was evaluated using accuracy, per-class precision, recall, F1, macro F1, and Cohen’s Kappa. While macro-F1 was initially selected as the primary summary metric for its equal penalisation of failure on either class, sensitivity and Cohen’s Kappa emerged as the most informative metrics when analysing the observed results. A brief image-level analysis is also reported in Section 4.5. All scripts used in this study are available under this repository [14].

3.7 Computational Costs & Sustainability

Unlike prior work reviewed, this study also documents computational costs and sustainability considerations. Token consumption and cost figures across all 36 experimental conditions are summarised in Table 2. GPT-4o-mini was by far the most expensive model, both in cost and token usage, consuming approximately $17\times$ more tokens than Qwen3.5 and $24\times$ more than Gemini 2.5 Flash Lite. This disparity is primarily attributable to OpenAI’s vision tile system, which decomposes images into fixed 512x512 pixel tiles before processing, significantly inflating input token counts relative to the other two models. Gemini 2.5 Flash Lite was effectively cost-free for this study, consuming only approximately 17 DKK of available free trial credits, while Qwen3.5 required a \$10 deposit to unlock free credit access, with actual inference costs remaining minimal. As discussed in Section 3.4, the smaller models were preferred in order to reduce the total energy footprint of this extended experiment.

Model	Provider	Total Tokens	Cost
GPT-4o-mini	OpenAI	853,249,054	\$64.29
Qwen-3.5-9B	TogetherAI	49,424,617	~\$10
Gemini-2.5-flash-lite	Vertex AI	34,650,234	~17 DKK

Table 2: Total token consumption and estimated cost per model across all 36 experimental conditions.

4 Results

4.1 Deepfake Detection Performance

Figure 4 presents the Macro-F1 scores across all six prompting strategies, models, and languages. In English, all three models stay above the 50% random-chance baseline throughout the entire experiment, with Macro-F1 scores ranging from 62.2% (GPT, S4) to 85.3% (Gemini, S6). In Romanian, GPT falls below random choice at S1, with 45.9% score, while, under the same prompt and strategy, Qwen achieves the highest Macro-F1 across the entire study with roughly 88%. The plot concludes that across both languages, S6, with an overall average Macro-F1 score of 80% across all models, yields the highest Macro F1 for the majority of conditions, with values ranging from 75.3% (Qwen, RO) to 85.3% (Gemini, EN). Strategies S2 to S5 produce broadly similar results, with average Macro-F1 score falling under the range 64.5% to 67%, forming a slight decline relative to S1, 72%, before the sharp recovery under few-shot prompting conditions.

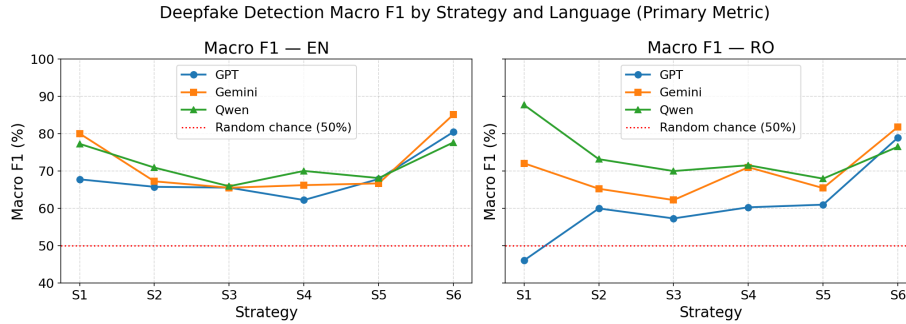


Figure 4: Macro F1 scores across prompting strategies S1–S6 for all three models, shown separately for English (left) and Romanian (right).

S1 = simple prompt only including the image

S2 = simple prompt that also includes the caption associated with the image on social media

S3 = C-o-T prompt. **S4** = Persona-building prompt

S5 = C-o-T + Persona. **S6** = Few-shot prompt with 4 example image + caption associations

The red dotted line indicates random chance performance (50%).

Figure 5 shows the corresponding accuracy values, which follow the same general pattern as Macro-F1. The close alignment between the two metrics is due to the balanced class distribution (600 real, 600 fake), which limits the

divergence between accuracy and Macro-F1 under symmetric misclassification. Accuracy ranges from 56% (GPT, S1, RO) to 85% (Gemini,S6, EN) and 88% (Qwen, S1, RO). However, aggregate accuracy conceals an underlying asymmetry between the models’ ability to detect fake images and their ability to clear real ones, which is examined in detail in Section 4.2 through sensitivity and specificity analysis.

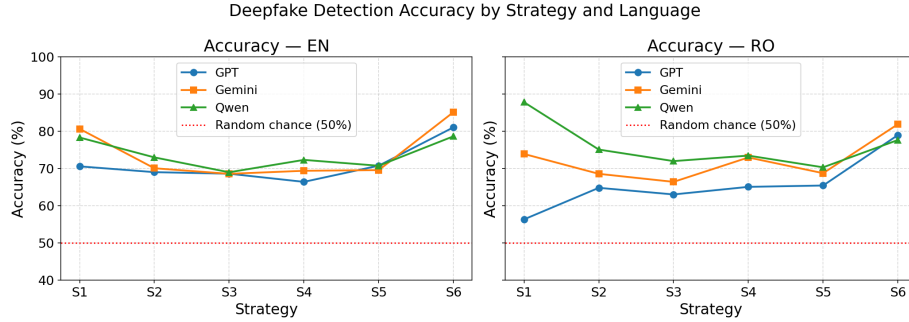


Figure 5: Accuracy scores across prompting strategies S1–S6 for all three models, shown separately for English (left) and Romanian (right). The red dotted line indicates random chance performance (50%).

Table 3 reports Cohen’s Kappa between each model’s predictions and the author’s annotations. Using the same Landis and Koch [26] scale, most conditions fall within the fair (0.21–0.40) to moderate (0.41–0.60) range. The single highest Kappa is recorded for Qwen-RO-S1, $\kappa = 0.757$, indicating a substantial agreement, while the lowest belongs to GPT-RO-S1, $\kappa = 0.127$, marked as slight agreement. At S6, all model–language combinations reach at least moderate agreement, with Gemini-EN-S6 achieving the highest S6 value, $\kappa = 0.703$. Cohen’s Kappa for S3 is consistently lower than for S1 across all models and languages, indicating that the Chain-of-Thought strategy does not improve agreement with ground truth relative to the simple baseline.

Full results for Accuracy, Precision, Recall, and Macro F1 are provided in Appendix D (Tables 4–7).

Table 3: Cohen’s Kappa between model predictions and ground truth per condition.

Model	Lang	S1	S2	S3	S4	S5	S6
GPT	EN	0.412	0.380	0.377	0.328	0.411	0.622
	RO	0.127	0.298	0.262	0.302	0.310	0.580
Gemini	EN	0.613	0.402	0.372	0.387	0.392	0.703
	RO	0.478	0.372	0.328	0.460	0.375	0.638
Qwen	EN	0.566	0.460	0.380	0.447	0.415	0.573
	RO	0.757	0.528	0.440	0.470	0.407	0.553

4.2 Sensitivity, Specificity and Cross-Lingual Performance

Figure 6 presents sensitivity, the proportion of AI-generated images correctly flagged across the six prompting strategies and for both languages.

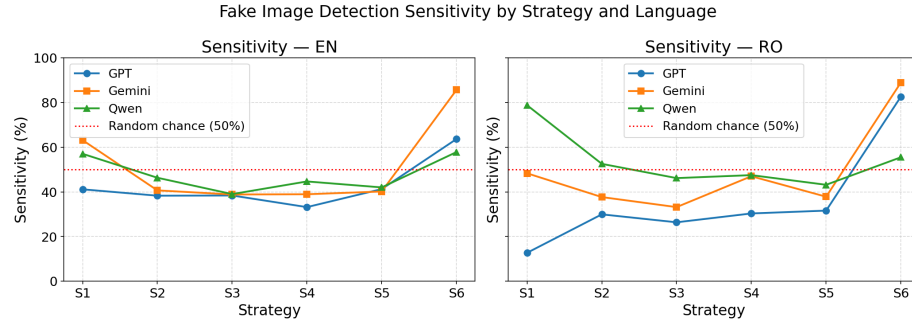


Figure 6: Sensitivity (or TPR) measured across prompting strategies S1-S6 for all three models, shown separately for English (left) and Romanian (right). The red dotted line indicates random chance performance (50%).

All three models begin with moderate sensitivity at S1 in English: GPT 41%, Gemini 63%, and Qwen 57%. Similarly, they all remain relatively flat between S2-S5 and considerably increase in S6, with a 20% increase for GPT and Gemini compared to S1, while Qwen’s sensitivity barely increases, at 58%. When the prompt is written in Romanian, the models present more variation across the experiment. GPT records only 13% at S1, the lowest value measured across the entire experiment. Gemini starts at 49% and Qwen at 79%. Across S2–S5, no systematic improvement is observed for either GPT or Gemini, mirroring the English pattern. Qwen’s high S1 sensitivity does not persist and declines across the intermediate strategies, before partially recovering at S6. By S6, GPT and Gemini record their highest Romanian sensitivity values: GPT 82%, Gemini

88%, while Qwen sits at 55% barely above chance. This reinforces S6 as the most effective strategy for fake detection regardless of prompt language, when it comes to language-first models.

Specificity, the proportion of real images correctly marked as real, is near-perfect across all conditions and models, between 95% and 100%, regardless of prompting strategy or language. Given these near-perfect results, no plot was worth presenting. This pattern is noted here for completeness and will be discussed in relation to the authenticity bias observed across models in the Results section.

Figure 7 shows the EN vs RO decision consistency heatmap, representing the percentage of images for which each model produces the same verdict (irrespective of correctness) across the two prompt languages. Consistency values range from 77.4% (GPT, S6) to 96.5% (Gemini, S2). The Chain-of-Thought prompt produces a modest but consistent dip in cross-lingual agreement across all models, with 90.2% for GPT, 86.5% for Gemini, and 87.7% for Qwen. At S6, Qwen presents the highest cross-lingual stability at 95.3%, while GPT records the lowest value at 77.4%, corresponding to 272 images where the verdict differed between languages. The disparity between high cross-lingual consistency and the substantial sensitivity degradation in Romanian observed in Figure 6 is addressed in Section 5.3.

EN ↔ RO Decision Consistency (%) — same image, same verdict across languages?

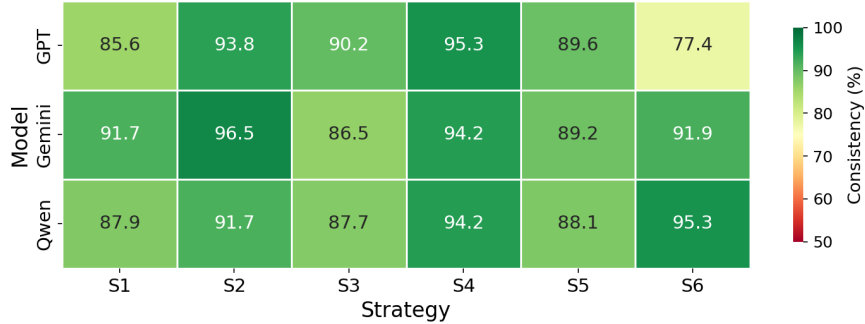


Figure 7: Deepfake decision Consistency measured across prompting strategies S1–S6 for all three models.

4.3 Propaganda, Manipulation Detection and Political Preference

Across strategies S2–S6 and both languages, all three models flag the majority of images as containing political propaganda, though flagging rates vary considerably by strategy and model. The GPT model responses range from 64.3% (RO S3) to 86.5% (EN S4). Gemini follows a similar pattern but diverges sharply at S6, where its flagging rate drops to 51.8% in English and 57.2% in Romanian, the lowest values recorded across all conditions. Qwen displays the opposite trajectory at S6, reaching its peak of 90.1% in English. Across all models, S4 (Persona) consistently produces the highest propaganda detection rates, while S3 (Chain-of-Thought) reduces flagging relative to the S2 baseline in every model-language combination, suggesting that explicit reasoning steps make models more conservative in their propaganda judgments. The full propaganda flagging rates per condition are provided in Appendix D, table 8. Figure 8 presents the distribution of manipulation techniques identified across all conditions, expressed as a percentage of propaganda-flagged responses and aggregated across all models, strategies, and languages. As multiple techniques can be assigned per image, the values sum up to more than 100%. Emotional appeal (Technique 1) is by far the most frequently identified technique, appearing in 71.8% of propaganda-flagged responses, indicating that models broadly associate political imagery with affective persuasion, and suggesting that emotional appeal is one of the most commonly used manipulation techniques used on Romanian Social Media. Simplification (Technique 5) and false context or decontextualization (Technique 3) rank second and third at 46.1% and 42.8%, respectively, while name-calling and smear (Technique 2) and appeal to authority (Technique 4) are less commonly invoked at 35% and 28%.

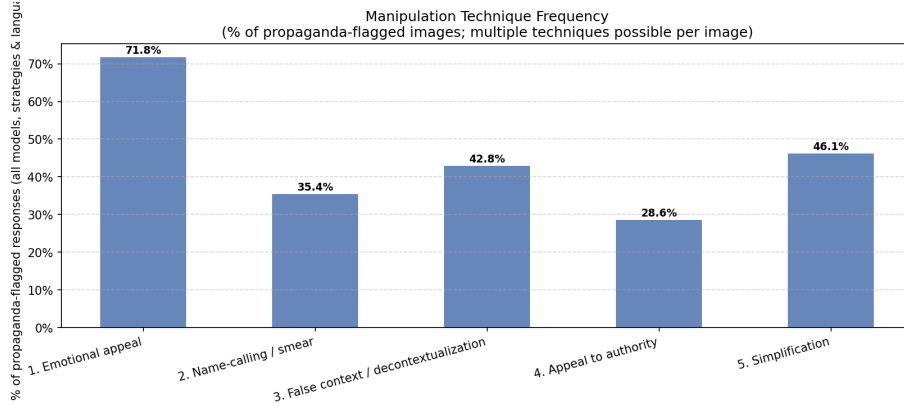


Figure 8: % of propaganda-flagged responses tagged with each technique (1–5), aggregated across all models, strategies, and languages. Multiple techniques allowed/image.

Models assign a right-leaning political preference far more frequently than left-leaning across all strategies. Aggregated over S2–S6, Qwen assigns right-leaning labels the most, at an average of 53.5% in both languages, while GPT assigns right in 28.2% of English responses, rising to 37.5% in Romanian. Gemini is the most conservative, assigning right in only 23.1% of English and 24.4% of Romanian responses, and predominantly staying Neutral: 73.7% EN and 74.3% RO. Left-leaning assignments are marginal across all models: GPT records the highest left rate at 15.3% in English, while Gemini assigns left to only 3.2% of English and 1.3% of Romanian responses. The dominance of right-leaning over left-leaning labels is consistent with the dataset’s composition, in which the majority of fake images originate from right-wing political pages or right-wing supporters. This also aligns with the remark about Romania missing a prominent left-wing political party. The full per-strategy breakdown and aggregated distribution across languages are provided in Appendix D, Figure 19.

Examples can be seen in figure 9.



(a) AI-generated image flagged as fake and propaganda 30/36 times,

Caption (translated): *"In Giurgiu, a county capital, a European city, in the year 2026, winter is KILLING people! And this is happening under the PNL (Liberal National Party) administration, with the Giurgiu City Hall led by PNL. For 3 days, local authorities were warned about today's weather phenomenon and took absolutely no action-they did nothing to prevent what happened."*

Manipulation techniques flagged consistently: 1. emotional appeal, 2. name-calling or smear, 5.simplification. '



(b) AI-generated image flagged as fake and propaganda 30/36 times,

Caption (translated): *"catârubd Next"*

Manipulation techniques flagged consistently: 1. emotional appeal, 3.false context or decontextualisation. 2. name-calling or smear.'

Figure 9: Examples of image & caption associations flagged as propaganda/manipulation.

4.4 Visual Cue Analysis

Figure 10 presents the frequency of visual cues used by the LLMs to motivate their decision of marking an image as not authentic, for three prompting strategies, S2(Simple question, using caption and the image as the only guidance), S4 (Persona), and S3/S5 (CoT), expressed as a percentage of the total images marked as fake, aggregated across all models and languages. As multiple cues can be assigned per image, values do not sum to 100%.

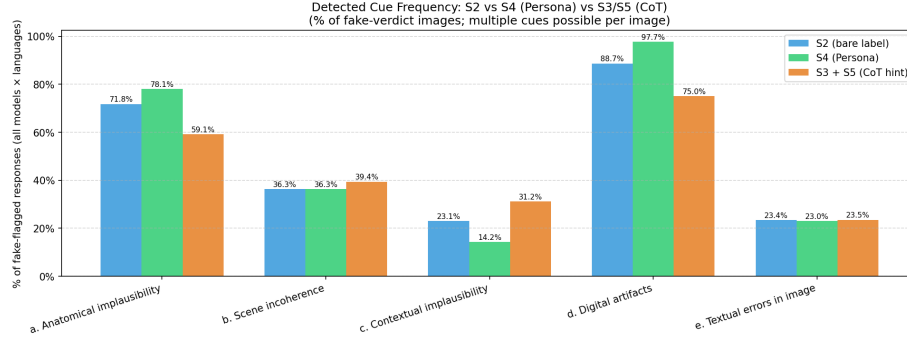


Figure 10: Detected visual cue frequency across fake-flagged images by prompting strategy, aggregated across all models and languages.

The plot shows that, irrespective of the prompting strategy, the visual cues used by LLMs are pretty much the same. Digital artefacts (cue d) is the dominant cue across all three strategy groups, detected in 88.7% of S2 responses, 97.7% of S4 responses, and 75.0% of S3/S5 responses. Anatomical implausibility (cue a) is the second most frequently selected cue, appearing in 71.8%. Contextual implausibility (cue c) shows the largest variation across strategies, while S2 and S3/S5 report 23.1% and 31.2% respectively. S4 drops sharply to 14.2%, suggesting that the Persona framing suppresses contextual reasoning in favour of low-level visual inspection. Textual errors in image (cue e) remains stable across all three groups at approximately 23%, indicating that the explicit diacritic hint introduced in S3/S5 does not meaningfully increase the detection of textual cues beyond the baseline. However, when broken down by language, textual errors are detected in 19.6% of Romanian-prompted fake-flagged responses compared to 10.7% in English, a difference explored further in Section 5.3. Word clouds of the reasoning text, filtered to fake-verdict images and presented separately for English and Romanian, are included in Appendix D, figure 20. They offer a qualitative complement to the cue frequency chart, illustrating the vocabulary models use when describing visual manipulation, but its analysis is out of scope for this research.

Examples are provided under figure 11.



(a) AI-generated image flagged as fake and propaganda 28/36 times,
Caption (translated): "Touching incident: an elderly woman from Tehran shot down an American drone with a jar of hummus imported from Israel."

Visual cues flagged consistently: b.Scene Incoherence, c.Contextual implausibility.



(b) AI-generated image flagged as fake and propaganda 31/36 times,
Caption (translated): "Romania woke up!!"

Visual cues flagged consistently: d.Digital artefacts, e.Textual errors in image. '

Figure 11: Examples of image & caption frequently flagged under the same visual cues.

4.5 Image-Level Analysis

Figure 12 presents the distribution of per-image disagreement scores across all 1,200 images, aggregated over all models, strategies, and languages. The disagreement score ranges from 0 (unanimous verdict across all 36 experiments) to 1.0 (perfectly split between fake and real, 18 times marked as real, 18 times marked as fake). Real images cluster near zero, with a mean disagreement score of 0.05, indicating near-universal consensus that real images are real, which matches the observations made about specificity. Fake images are distributed across the full range, with a mean of 0.43 (corresponding to a 27/9 split in

the verdict across the experiments), reflecting substantially greater variability in model verdicts. Of the 9 images that received a perfectly split score of 1.0, all are fake. Similarly, 86 of the 88 images with a disagreement score above 0.8 are fake, while 576 of the 744 near-consensus images (score below 0.2) are real. These patterns reinforce the authenticity bias observed in the sensitivity analysis, as models converge on real verdicts far more reliably than on fake ones, with the hardest images being almost exclusively within the fake set. Examples can be seen in the appendix **D**, figure **21**.

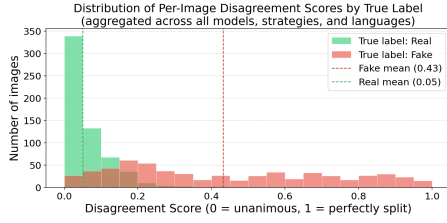


Figure 12: Distribution of per-image disagreement scores by true label, aggregated across all models, strategies, and languages.

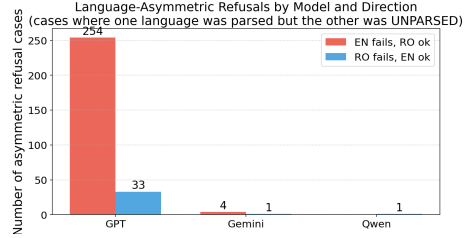


Figure 13: Number of language-asymmetric refusal cases by model and direction, aggregated across all strategies.

Figure **13** presents the count of language-asymmetric refusals, specific cases where one language produced a valid response while the other was unparsed, either a refusal, an interdiction met by the model, or something similar. A total of 293 such cases were identified across all models and strategies, of which 287 (98.0%) stem from the GPT model. The most common scenario is English prompt failing and Romanian prompt working, accounting for 254 of GPT’s 287 asymmetric cases, compared to only 33 in the reverse direction. Gemini and Qwen contribute negligibly, with 5 and 1 cases, respectively. The asymmetric refusals are distributed roughly equally between fake and real images (156 fake, 137 real), indicating no systematic bias toward refusing images from either class.

5 Discussion

5.1 Effect of Prompt Strategy

The results reveal that the prompting strategy has a decisive effect on deep-fake detection performance, but the results may still be considered surprising. Strategies like Chain-of-Thought (S3), Persona (S4), or even their combination

(S5) do not produce improvements over the simple prompt (S1) in the majority of experiments, but rather decrease performance. Few-shot prompting (S6) is the strategy that produces better results across all models and languages. This builds on the narrative that more complex prompts do not always lead to better, more trustworthy results.

The Sensitivity and Specificity reveal the true differences across the prompting strategies used during this research. Specificity falls, across the entire 36 conditions in the range 95%-100%, whereas sensitivity varies substantially across strategies and models. Figure 6 shows that between S2-S5, sensitivity falls below the 50% random chance threshold for all models and languages, except for Qwen S2 RO, meaning that, without few-shot guidance, the LLMs perform worse than random at classifying images as fake. This reflects a strong authenticity bias that the LLMs likely inherited during training. The examples provided during S6 successfully overcome this bias, as the language-first models, GPT and Gemini, reach their highest sensitivity across all conditions, GPT 64% in English and 82% RO (compared to just 13% in S1 RO, marking a significant 69 point increase), Gemini 85% EN and 88% RO. The visual-first model Qwen also improves at S6 compared to S2-S5, but its best sensitivity is observed at S1, suggesting it relies primarily on visual features before textual context is introduced. The near-perfect specificity confirms that no strategy causes models to over-flag real images, the entire performance gap between strategies is driven by their ability, or inability, to catch fakes. The high specificity scores may also partly reflect the inherent quality difference between the two classes: real images are authentic photographs with natural coherence, high in quality, taken by professional photographers at political events, while fake images, despite being convincing to the human eye, may carry subtle low-level artefacts that reinforce the model’s default assumption of authenticity, making them harder to confidently flag as fake.

Table 3 further solidifies these observations. The few-shot strategy is the only strategy where all model-language combinations reach at least moderate agreement with ground truth $\kappa = 0.56-0.70$, with half of the conditions reaching substantial agreement. S2 to S5 perform worse compared to S1 for every model-language combination. The single worst condition in the entire table is GPT-S1-RO ($\kappa = 0.127$), which sits in the slight agreement bracket, consistent with its near-zero sensitivity of 13%, performing worse than random. The largest single cross-lingual gap in the same model and strategy appears for GPT-S1, as $\kappa = 0.41$ in English, resulting in a 0.28-point compared to the Romanian

agreement score, a shift driven solely by the language of the prompt, with the image and task identical in both conditions. The Macro-F1 scores, captured in Figure 4, also reflect the same pattern, where S6 produces the highest score in the majority of conditions.

The convergence at S6 is the clearest finding of the study. Showing a model four labelled examples of what fake and real images look like is more effective than instructing it to reason step by step or to adopt the persona of an expert in deepfake detection. The examples provide concrete visual anchors that abstract instructions cannot replicate. The underperformance of S3 is particularly notable given previous research documenting Chain-of-Thought as an effective reasoning enhancement, suggesting that the benefits of step-by-step reasoning do not transfer straightforwardly to visual detection tasks.

5.2 Cross-Model Comparison

The three models used for this study have two distinct architectural structures: GPT-4o-mini and Gemini-2.5-flash-lite are language-first models, trained first on text with vision capabilities later added, whereas Qwen-3.5-9B is a visual-first model, with a dedicated visual encoder. This distinction can also be observed in the model’s behaviour across the experiment.

The metrics observed for the language-first models follow very similar trajectories, with sensitivity flat or declining between S1 and S5, with a sharp improvement at S6. Their performance is largely language dependent, as both GPT and Gemini degrade in Romanian relative to English across most strategies. On the contrary, Qwen’s highest sensitivity is observed at S1, 79% in Romanian and 57% in English, with constant decreases occurring between S2 and S5, before achieving a slight recovery in S6, 55% in Romanian and 58% in English. This suggests that Qwen’s visual encoder capabilities are best used when operating on the image alone, without textual context or a complex prompt. The introduction of Romanian post captions in S2 and beyond appears to shift Qwen toward language-driven reasoning, where its Romanian capability is weaker, suppressing the visual capabilities that led to strong results at S1. The most complex prompt, S5, combining Chain-of-Thought reasoning and Persona, leads to Qwen’s lowest Sensitivity score in Romanian, at 42%.

OpenAI’s model accounts for 287 of the total 293 language-asymmetric refusal cases that occurred in the study, where one language produced a valid response, and the other did not. 254 of those failed in English and returned a

valid response in the Romanian prompt, suggesting that GPT’s content moderation is more sensitive to explicit political language in English rather than other low-resource languages such as Romanian. Gemini and Qwen, with 5 and 1 cases each, indicate the significantly different filtering systems applied by different LLM providers. The topic is further discussed in section 5.5.

A further matter that must be considered is the models’ limited familiarity with Romanian political figures. LLMs are trained predominantly on English-language data and are therefore likely more exposed to globally prominent politicians, whose faces, activity, and political contexts are better documented across the web, making it easier to assess whether an image is plausible given known real-world events. Romanian political figures, by contrast, appear far less frequently in pre-training corpora, thereby reducing the model’s ability to cross-reference images with known appearances or political contexts. As a result, part of the detection signal in some conditions may come from contextual implausibility at a global level rather than purely visual forensic reasoning, and using this framework may yield better results on a dataset composed of deepfakes and authentic images of more internationally prominent political figures.

During the first phases of research for this project, an interesting observation has been made: Gemini models refuse to generate fake images depicting real political figures (either when asked by text or when a starting image is provided), while GPT models have no such restriction. Even though a further dive into this potential policy difference is outside of the scope of this research, the introduction of such restrictions could meaningfully limit the production of AI-generated political propaganda at scale, a development that would have direct positive implications for the integrity of political discourse on social media for the long term.

5.3 Cross-Lingual Performance

The EN vs RO decision consistency heatmap, figure 7, shows that overall consistency, where models produce the same verdict (irrespective of correctness) across the two prompt languages, is high across all conditions, ranging from 77% (GPT,S6) to 97% (Gemini,S2). At first, this suggests that prompt language has little to no effect on model behaviour, with images receiving the same verdict regardless of it. This interpretation is, however, quite misleading. Consistency measures whether the model gives the same verdict in both languages, not whether the verdict is correct. A model that predicts Real for almost ev-

ery image will achieve high consistency while performing poorly. This scenario also yields decent accuracy, which is why accuracy is not really relevant for this research. The degradation caused by language manifest is sensitivity and Macro-F1, not in consistency, which is why all of those are necessary to measure cross-lingual performance.

GPT’s results at S1 are the best illustration of this. It achieves an 85.6% consistency, while sensitivity in Romanian is at the lowest measured across the experiment, at 13%, as the model predicts Real for the vast majority of images in both languages. The verdicts, therefore, are identical (hence great consistency), but both languages lead to largely mistaken results for the fake set of images. The language of the prompt shifts how much the model defaults to classifying images as authentic. GPT under the S6 prompt records the lowest consistency, at 77%, meaning 272 images out of the total of 1200 received different verdicts when the language was changed. This suggests that the introduction of few-shot examples, while improving overall sensitivity, also introduces language sensitivity. The examples anchor the model’s decision boundary differently depending on the language context in which they are presented.

Overall, performance degrades in Romanian relative to English for GPT and Gemini across all the conditions bar S4 for Gemini, while Qwen shows the opposite pattern, performing best in Romanian across all strategies except S6.

5.4 Romanian-Specific Findings

The most domain-specific finding of this study concerns the detection of textual errors within images, cue (e) in the visual cue framework. During the collection of data, and during the past years, the author observed that AI-generated images are usually either entirely missing Romanian diacritics (ă, â, î, ș, ț), or they appear partial, or with malformed characters on top of or below the corresponding letter. This discovery is strongly supported by data, as the cue (e) is flagged almost twice as much in images identified as fakes during the Romanian prompt (1067/5444, almost 20%), compared to the English prompt (642/6020, 11%). This difference is not driven by prompting strategy, as the distribution of visual cues identified follows the same pattern in S2, S4, and the combination of the C-o-T driven strategies, S3 + S5, as can be seen in Figure 10. This is a strong indicator that the explicit diacritic hint introduced in the C-o-T prompt does not meaningfully amplify detection beyond what the Romanian language context alone achieves. The prompt language itself, by directing the model’s

attention toward Romanian text, appears sufficient to heighten sensitivity to diacritic errors in the image. Examples of images with faulty diacritics can be seen in figure 14.



(a) AI-generated image with malformed Romanian diacritics visible in the overlaid text. 'â' instead of 'ă'. Unclear sign on top of 'a' on the second row, seemingly a combination of 'â' and 'ă'



(b) AI-generated image with malformed Romanian diacritics in the overlaid text. 'ș' also has a comma on top.

Figure 14: Examples of diacritic errors in AI-generated images from the dataset. Characters such as ș, ț, ă, â, and î are absent or malformed, a recurring artefact of current image generation models when producing Romanian-language text.

This finding has no direct equivalent in the English-language deepfake detection literature, given the lack of special characters, and textual errors in images are rarely an indicator of AI-generated images. For Romanian political content specifically, malformed diacritics are a reliable, visually accessible indicator of AI-generated content, and are immediately identifiable to any native, expert, or non-expert speaker. It is one of the clearest original contributions of this study.

5.5 Model Refusals and Format Compliance

Both format compliance and content moderation behaviour differed substantially across models throughout the experiment. The GPT model presented the most significant challenges in terms of refusals, as also mentioned in previous sections. During the testing of the English Chain-of-Thought prompt, GPT refused to answer in 4 out of the 10 provided images, due to explicit political manipulation language provided in the reasoning steps. A more moderated prompt was produced by softening the phrasing, resolving the issue, and bringing the refusal rate from 40% to 8.7%. No equivalent moderation triggering was observed for Gemini and Qwen under any condition, which solidifies the finding in Section 5.2 that GPT’s content moderation is more sensitive to explicit political language in English.

This pattern is reflected in the counts of skipped responses across the experiment. GPT-4o-mini recorded 28 and 29 skipped responses in S2 English and Romanian, respectively, rising sharply to 104 in S3 English and 160 in S5 English, the latter representing 13.3% of the entire dataset, proving that the political text included in the reasoning steps of the C-o-T strategy triggers some refusals. Gemini and Qwen recorded negligible skipped images/refusals across all conditions, with a maximum of 4 in any single condition.

Qwen presented distinct format compliance challenges in Romanian. Qwen-3.5-9B is a high-reasoning model with a built-in extended thinking mode, which was turned off to keep the output tokens to a reasonable number, increase speed, and adhere to the desired strict output format. This has been observed during the test in S1, where the model would use more than 2000 output tokens just to end up with a ‘Yes’/‘No’ answer. During the more complex prompts, the model produced unsolicited analytical commentary sections in Romanian appended to its structured output responses. This behaviour was absent in English prompts and was not observed in GPT or Gemini under any condition, highlighting a cross-model difference in instruction-following for structured output in lower-resource languages. Gemini and GPT were largely compliant with the required output format across all strategies.

5.6 Qualitative Case Studies

To complement the quantitative findings, this section examines a selection of individual images across the observed best strategies, S1 and S6. The images were selected to represent distinct challenge categories: collage content, subtle

visual artefacts, explicit authenticity markers, and ambiguous real images.

Image 154, **15a**, is a collage made out of a real picture (the one on the left) and a fake one. It also includes 'Pamflet/Satiră' (Pamphlet/Satire) written on it. Across S1 and S6, the image was correctly flagged as fake in 7/12 situations, with Gemini being wrong 2 times in S1 and Qwen 2 times in S6.

Image 17, **15b**, presents another interesting situation, in which the data presented is true, but the picture behind is a deepfake, observed by looking at hand and finger artefacts. This was flagged as fake only 4/12 times, highlighting the models' tendency to overlook subtle visual artefacts.

Image 340, **15c**, is a deepfake, as also marked on the image itself by the text: 'Created by: ChatGPT', in Romanian. This image is correctly flagged as fake 11/12 times, GPT-S1-RO being the only wrong verdict.

Image 1167, **15d**, the one with the highest contradiction across the real dataset, had 7/12 correct responses across S1+S6. The image is significantly lower in quality compared to the other real images collected for this study, which explains the indecision of the LLMs. The image is collected from the Facebook page of the Romanian Army.

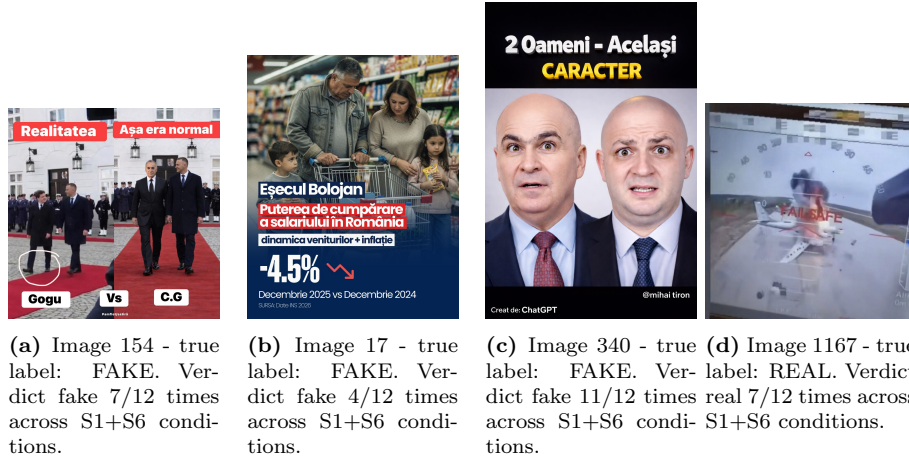


Figure 15: Case Studies.

6 Conclusion and Future Work

6.1 Summary of Findings

This research evaluated the ability of three consumer-grade large language models, GPT-4o-mini, Gemini-2.5-flash-lite and Qwen-3.5 Vision, to classify 1200 images collected from the noisy Romanian Social Media as either AI-generated/AI-altered or real. The experiment tested those models across six different prompting strategies and 2 languages, 1 high-resource (English) and 1 low-resource (Romanian), resulting in 36 different experimental conditions. The answer to the main research question, **1**, which questions the capacity of consumer-grade LLMs to detect deepfakes, is that, in their default configuration, these models cannot be reliably trusted for this purpose, particularly when applied to noisy, real-world social media content rather than controlled benchmark datasets.

RQ1 (see 1) - Detection accuracy. The main finding of this research is that few-shot prompting (S6) is the only strategy that overcomes the authenticity bias exhibited by all three models, as all 6 model-language combinations reach moderate to substantial agreement with the true labels, $\kappa = 0.56-0.7$. All models default to "Real" predictions across the majority of conditions and fall below random chance at detecting deepfakes without explicit guidance. These results suggest that consumer-grade LLMs cannot be considered reliable deepfake detectors, particularly when applied to noisy, creative, real-world social media content.

RQ2 (see 2) - Cross-lingual performance, RO-EN. Detection performance degrades significantly in Romanian for language-first models. The GPT model records the most severe performance degradation, from $\kappa = 0.41$ in English to $\kappa = 0.13$ under the zero-shot baseline, simple prompt, while Gemini degrades moderately but more stably across strategies. Qwen, by contrast, achieves its highest performance under a simple Romanian prompt, demonstrating that cross-lingual degradation is model-dependent rather than universal. ChatGPT remains, to this day, the most widely used AI assistant in Romania, yet querying it in Romanian without a structured prompt (S1 RO) produces the worst single condition in the entire study, performing significantly worse than a random choice, with a sensitivity of only 13%, meaning it correctly identifies fewer than one in eight deepfakes. Therefore, a Romanian-speaking non-expert user, relying on ChatGPT in its default conversational mode to assess whether

an image is AI-altered, is, in practice, effectively receiving no reliable guidance at all. For English, few-shot prompting with either the GPT or the Gemini model represents the most effective configuration, with both these language-first models reaching substantial Kappa scores at S6. For Romanian, Qwen 3.5 Vision under a simple prompt is the most accessible option for users who are not familiar with a few-shot prompt, while S6 in Romanian achieves decent performance across all three models.

RQ3 (see 3) - Prompt strategies. Few-shot prompting is the only strategy that consistently improves performance over the simple baseline prompt. Chain-of-Thought and Persona prompting do not produce reliable improvements and, in several conditions, lead to performance degradation. Providing two labelled real and two labelled fake image-caption pairs as examples outperforms explicit step-by-step reasoning instructions, suggesting that anchoring the model’s decision boundary with concrete examples is more effective than instructing it to reason carefully.

RQ4 (see 4) - Language-first vs Vision-first models. The language-first models achieve best performance under few-shot prompting conditions, where explicit examples compensate for the models’ tendency to rely on the caption and linguistic framing rather than visual artefacts. The vision-first model, Qwen 3.5 Vision, under simple baseline conditions, in Romanian, is the only exception to this pattern, where the vision-first model makes best use of its visual capacities, achieving an average Macro-F1 of 82%. For Romanian-speaking users who cannot construct a few-shot prompt, Qwen 3.5 Vision under a simple prompt is therefore the most accessible and reliable option, while S6 in Romanian achieves competitive performance across all three models.

RQ5 (see 5) - Visual cues. Digital Artefacts and Anatomical implausibility are the most frequently invoked detection cues across all models and strategies. A domain-specific finding of this study is the systematic misrepresentation of Romanian diacritics in AI-generated content. AI-generated images consistently fail to reproduce Romanian diacritics (ă, â, î, ș, ț) accurately. Models prompted in Romanian invoked textual errors as a detection cue almost two times more than when prompted in English (20% of fake-flagged RO responses vs 11% in EN). This signal has not been documented in any prior deepfake detection research.

RQ6 (see 6) - Practical guidelines. Based on the findings above and the observations made during the experiment, empirically grounded recommendations for detecting AI-generated images and identifying political manipulation

are provided in section 6.4.

The full prompts, as well as the scripts used during this research, can be found under this Github Repository [14].

6.2 Limitations

Despite the interesting findings discussed above, it must be acknowledged that this research has several limitations. The dataset of 1,200 images was manually annotated solely by the author due to time constraints, which may introduce bias, particularly given the author’s inherent political preferences and prior familiarity with Romanian social media. Inter-annotator agreement was assessed on a subset of 100 images with 2 independent non-expert annotators as time-restricted full annotation. The fake set is composed of more right-wing and sovereignist content, reflecting the real-world distribution of AI-generated political digital content on Romanian social media rather than a collection bias, while also taking into consideration the absence of a prominent left-wing political party. Another limitation is that the dataset used in this study cannot be made publicly available, as publishing images collected from Facebook is restricted by the platform’s Terms of Service [30], the right to erasure, and applicable GDPR regulations. Full reproducibility may therefore be limited, as some posts may have been deleted since collection. An index of post URLs and the author’s annotated labels are, however, publicly available on Zenodo [13].

Propaganda and manipulation technique annotations were only produced by the models, rather than by any human annotators. The outputs should, therefore, be interpreted as general recommendations rather than ground truth.

The results may also be partially explained by the image quality differences between the two sets: authentic images, often captured by professional photographers, tend to exhibit higher resolution and quality, which may inflate specificity independently of the models’ visual reasoning capacity.

Budget and time limitations prevented testing more capable and newer models, such as GPT-4o or Gemini 2.5 Pro. These models may exhibit better visual capacities and may yield better results. This research also fails to analyze potential model output volatility, as each image has been evaluated only once per condition. Multiple runs of the same image under the same combination of model, prompting strategy, and language would also enable detecting the confidence behind the model’s output.

6.3 Future Work

Several possible directions emerge from the findings and limitations of this research.

The current 1,200-image dataset, annotated by a single researcher, would benefit from full independent annotation by trained human annotators, while also including labeling of propaganda and manipulation techniques. A larger and more ideologically diverse dataset, collected across multiple political contexts and languages, would strengthen the generalisability of the findings. The development of a separate dataset, capturing deepfakes and authentic images of global political figures, instead of Romanian political actors, would also lead to an interesting study. Given the motivation behind this research, an adjacent study looking into the LLMs’ capabilities of detecting fake accounts or comments left by bots should be considered.

The experiment, while extensive, could be further adjusted to provide even more interesting insight into LLM’s deepfake detection capabilities. The repeated run of the baseline and the best overall strategy, few-shot prompting, would allow for quantification of output volatility and provide a measure of prediction confidence absent from the current evaluation. A more in-depth image-level analysis, examining which specific images consistently divide model verdicts, and whether disagreement correlates with any specific image characteristics, would complement the aggregate metrics reported here. Image pre-processing to remove different elements overlaid on the image, such as ‘Pamphlet/Satiră’ labels, platform-generated AI disclosure icons, or the text, would isolate genuine visual reasoning from label reading. Including the name or the known political preference of the posting account as a contextual signal could also lead to interesting findings.

According to the author’s literature review, no prior work has investigated why AI-generated images usually contain unnatural colour palettes, overly saturated, unusual, and inconsistent with natural photographic colour distributions, despite this being a commonly reported perception among people encountering deepfake content on a frequent basis. Future work could examine where this difference stems from and how these colour palettes could be used as detection signals.

Finally, several technical details could be further developed and analyzed. Evaluation of newer and more capable models would extend the comparative analysis beyond the models tested during this study. Including models from

other providers, such as Grok or Deepseek, with known political preferences/-filters, could also make this research more interesting.

6.4 Practical Recommendations

The practical recommendations presented below are also available in Romanian under this GitHub Repository [14].

6.4.1 Visual Indicators of AI-generated/AI-altered Content

When assessing whether an image is real or AI-generated/AI-altered, pay close attention to:

1. **Digital artefacts.** Look for unnatural blurring, rendering errors.
2. **Anatomical implausibilities.** Examine facial features, skin texture, and hair. Pay close attention to the hands, the number of fingers, the shape, and the potential unnatural position of the fingers. Assess whether the skin colour of the hands matches the facial colour and the age range of the person present in the image.
3. **Scene incoherence.** Assess whether lighting, shadows, and perspective are coherent throughout the image. Look at the background objects, make sure they are not distorted or inconsistent with the main subject of the image.
4. **Contextual implausibility.** Recognise, if possible, the people represented in the picture. If they are political actors, search for documented political activity during the posting date of the image you are analysing. Use common sense to assess whether the image could be real.
5. **Textual errors.** Romanian text, diacritic errors, or missing characters (ă, â, î, ș, ț) are a strong indicator of AI-altered content. Analyse whether the special signs (˘, ^) called breve, circumflex, and comma below, are distorted, appearing in a different font, rendered only partially, or associated with the wrong letters. More broadly, any text within an image that appears blurred, garbled, or inconsistent in font and spacing should raise suspicion.
6. **Colour schemes.** Does the image present unnatural colour palettes? Do the colours seem ultra-saturated?



Ultra-saturated colours

Ultra-saturated colours

Unnatural colour palette

Figure 16: Example of AI-generated/AI-altered images with unnatural colour palettes or with ultra-saturated colours.

Furthermore, for users seeking automated assistance, if possible, use a vision-first LLM, such as Qwen-3.5-9B, accessible at <https://qwen.ai/home>. When using language-first models, such as those provided by OpenAI or Google, provide 2 labelled examples of authentic content from a trustworthy source, and 2 labelled examples of AI-altered content.

6.4.2 Common indicators of political manipulation

The following have been observed as common manipulation techniques in political content associated with AI-generated/AI-altered content:

1. **Emotional appeal.** Does the association of the image and accompanying text use fear, anger, or pride to bypass rational thinking? Is the post trying to trigger an emotional response, such as a strong sense of patriotism?
2. **Simplification.** A complex issue is reduced to a single villain. Certain details or nuances are stripped away to present one-sided narratives as obvious truths. Political situations are usually complicated.
3. **False context or decontextualization.** Even authentic images can be used to drive manipulation. Assess whether the image is being used in a context unrelated to its original meaning. Search for the original source of the image to identify the real context.

4. **Name-calling or smear.** Are people with a specific political orientation, or belonging to a specific group or political party, labelled in a certain way? Is a mockery attempted?
5. **Appeal to authority.** Do the image and the caption invoke symbols of authority, such as institutional logos, uniforms, official settings, or prominent figures, to make a claim credible? Verify whether the depicted authority figure or institution has genuinely endorsed the message being communicated by checking news and official social media pages.

AI Usage Disclosure

Given the nature and the scope of this BSc Project, AI has been used to assess how consumer-grade vision-language models can detect AI-generated content, as described throughout this thesis. Beyond the use of AI as a central subject of this study, the following AI-based tools have been used:

- **Claude** (provided by Anthropic) has been used for plotting, regex-matching Excel files generated from model outputs, and fixing LaTeX-specific formatting issues, such as Image insertion.
- **Grammarly**, already integrated into Overleaf, has been used, on the free tier, to fix grammar errors, misspellings, etc.
- **Google Scholar** AI Labs feature has been used in the beginning of the research to find relevant literature for the topic.

All the ideas, analyses, interpretations, and remarks made across this study belong in their entirety to the author.

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Appendix for
*"Deepfakes And Political Propaganda: A Multimodal
Study Of Deepfake Imagery And Accompanying Text"*
BSc Project by giur@itu.dk

A Dataset Examples

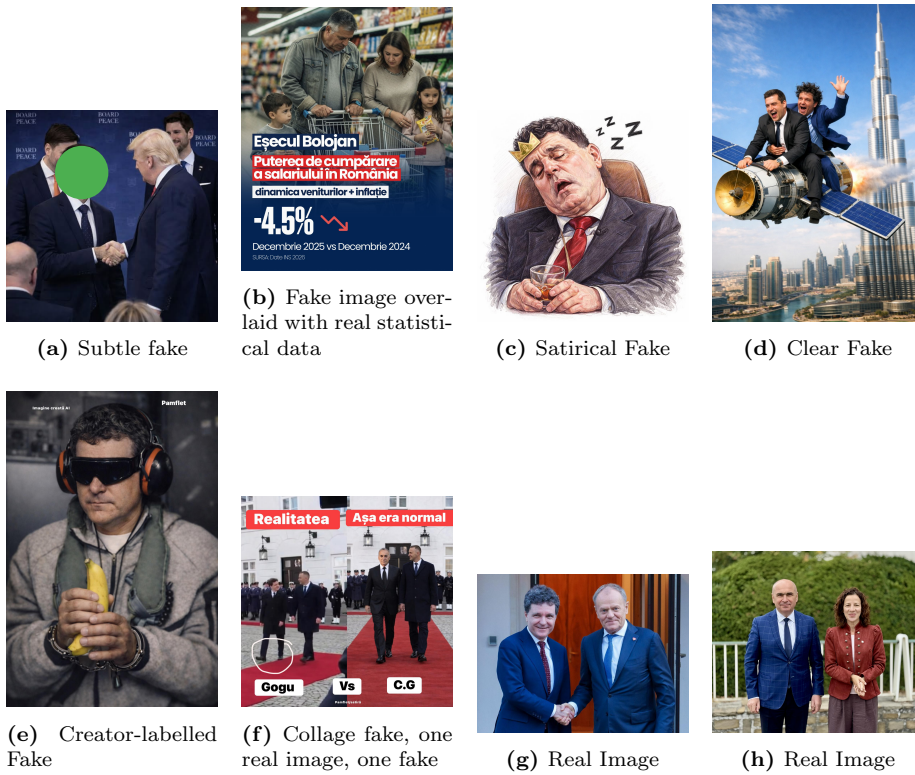


Figure 17: Representative dataset examples.

B Images provided as guidance in the Few-Shot prompt



ID: 23 — Type: Fake
 Field 1 (Image decision): Yes
 Field 2 (Detected cues): b, c, d
 Field 3 (Propaganda): Yes
 Field 4 (Technique): 2, 5
 Field 5 (Political preference): Right



ID: 640 — Type: Real
 Field 1 (Image decision): No
 Field 2 (Detected cues): N
 Field 3 (Propaganda): No
 Field 4 (Technique): N
 Field 5 (Political preference): Neutral



ID: 208 — Type: Fake
 Field 1 (Image decision): Yes
 Field 2 (Detected cues): a, c, d, e
 Field 3 (Propaganda): Yes
 Field 4 (Technique): 1, 4
 Field 5 (Political preference): Right



ID: 705 — Type: Real
 Field 1 (Image decision): No
 Field 2 (Detected cues): N
 Field 3 (Propaganda): Yes
 Field 4 (Technique): 1, 5
 Field 5 (Political preference): Right

Figure 18: S6 few-shot example images with their expected model outputs.

C Final Prompt Strategies

English prompt variants are provided below. Romanian variants follow the same structure and output format, with all field labels and instructions translated accordingly. Full Romanian final prompts, as well as other prompt variations, are available in the accompanying GitHub repository [14].

C.1 S1 – Simple

Determine if the attached image is AI-generated, AI-altered or Deepfake.

Required strict format:

Verdict: [Yes/No]

C.2 S2 – Simple with Context

Analyze the attached image and the associated context: {context}.

Required strict answering format:

1. Image decision: [Yes/No] (Is the image AI-generated/Deepfake/AI-altered?)

2. Reasoning — detected cues: [Select ALL that apply, comma-separated: a/b/c/d/e/N]

a. Anatomical implausibility / b. Scene incoherence / c. Contextual implausibility / d. Digital artefacts / e. Textual errors in image / N. Real image (no cues detected)

Select N only if image decision is No. If multiple cues are present, list all (e.g. a,d,e).

3. Political propaganda/manipulation: [Yes/No] (Can the post be considered political propaganda or manipulation?)

4. Manipulation/Propaganda technique: [Select ALL that apply, comma-separated: 1/2/3/4/5/N]

(1. Emotional appeal / 2. Name-calling or smear / 3. False context or decontextualization / 4. Appeal to authority / 5. Simplification / N. Not applicable)

Select N if field 3 is No. If multiple techniques apply, list all (e.g. 1,3).

5. Political preference: [Left/Right/Neutral]

C.3 S3 – Chain-of-Thought (moderation-safe variant, V4)

Analyze the attached image and the associated context: {context}.

Before answering, reason step by step through the following:

- Examine facial features, skin texture, hair, and hands for anatomical inconsistencies
- Look for digital artefacts, unnatural blurring, or rendering errors
- Assess whether lighting, shadows, and perspective are coherent throughout the image
- If the image contains text, check for missing, partial, or malformed diacritics (ă, â, î, ș, ț) or other textual errors
- Consider whether the image content is consistent with the provided context
- Consider the tone of the accompanying text and whether it appears neutral, charged, or polarising
- Assess whether the image includes institutional, authority-related, or symbolic visual elements and how they relate to the subject
- Consider whether the image content appears consistent with how the accompanying text presents it

Required strict answering format:

1. Step-by-step reasoning: [3–5 sentences summarising your observations above]
2. Image decision: [Yes/No] (Is the image AI-generated/Deepfake/AI-altered?)
3. Reasoning — detected cues: [Select ALL that apply, comma-separated: a/b/c/d/e/N]
 - a. Anatomical implausibility / b. Scene incoherence / c. Contextual implausibility / d. Digital artefacts / e. Textual errors in image / N. Real image (no cues detected)Select N only if image decision is No. If multiple cues are present, list all (e.g. a,d,e).
4. Political propaganda/manipulation: [Yes/No]
5. Manipulation/Propaganda technique: [Select ALL that apply, comma-separated: 1/2/3/4/5/N]
 - (1. Emotional appeal (uses fear, anger, or pride to bypass rational thinking) / 2. Name-calling or smear (attacks or demeans a person-

/group with labels) / 3. False context or decontextualization (real image or quote used in a misleading context) / 4. Appeal to authority (implies legitimacy through symbols, uniforms, or institutions) / 5. Simplification (complex issue reduced to a single villain or slogan) / N. Not applicable)

Select N if field 4 is No. If multiple techniques apply, list all (e.g. 1,3).

6. Political preference: [Left/Right/Neutral]

C.4 S4 – Persona

You are a professional digital forensics analyst specialising in detecting AI-generated images, deepfakes, and political misinformation on social media. You have extensive experience analysing manipulated visual content in Eastern European political contexts.

Analyze the attached image and the associated context: {context}.

Answer each field with the required code only. Do not include any additional text or notes. Respond with necessary codes only, no extra text.

Required strict answering format:

1. Image decision: [Yes/No] (Is the image AI-generated/Deepfake/AI-altered?)

2. Reasoning — detected cues: [Select ALL that apply, comma-separated: a/b/c/d/e/N]

a. Anatomical implausibility / b. Scene incoherence / c. Contextual implausibility / d. Digital artefacts / e. Textual errors in image / N. Real image (no cues detected)

Select N only if image decision is No. If multiple cues are present, list all (e.g. a,d,e).

3. Political propaganda/manipulation: [Yes/No]

4. Manipulation/Propaganda technique: [Select ALL that apply, comma-separated: 1/2/3/4/5/N]

(1. Emotional appeal / 2. Name-calling or smear / 3. False context or decontextualization / 4. Appeal to authority / 5. Simplification / N. Not applicable)

Select N if field 3 is No. If multiple techniques apply, list all (e.g.

1,3).

5. Political preference: [Left/Right/Neutral]

C.5 S5 – CoT with Persona

You are a professional digital forensics analyst specialising in detecting AI-generated images, deepfakes, and political misinformation on social media. You have extensive experience analysing manipulated visual content in Eastern European political contexts.

Analyze the attached image and the associated context: {context}.

Before answering, reason step by step through the following:

- Examine facial features, skin texture, hair, and hands for anatomical inconsistencies
- Look for digital artefacts, unnatural blurring, or rendering errors
- Assess whether lighting, shadows, and perspective are coherent throughout the image
- If the image contains text, check for missing, partial, or malformed diacritics (ă, â, î, ș, ț) or other textual errors
- Consider whether the image content is consistent with the provided context
- Consider whether the post uses emotional language, attacks individuals, or simplifies complex political issues
- Assess whether the image or context attempts to associate a person or group with authority symbols or institutions
- Consider whether the image or context could be misleading by presenting real content out of its original context

Required strict answering format:

1. Step-by-step reasoning: [3–5 sentences summarising your observations above]
2. Image decision: [Yes/No] (Is the image AI-generated/Deepfake/AI-altered?)
3. Reasoning — detected cues: [Select ALL that apply, comma-separated: a/b/c/d/e/N]
 - a. Anatomical implausibility / b. Scene incoherence / c. Contextual implausibility / d. Digital artefacts / e. Textual errors in image / N. Real image (no cues detected)

Select N only if image decision is No. If multiple cues are present, list all (e.g. a,d,e).

4. Political propaganda/manipulation: [Yes/No]

5. Manipulation/Propaganda technique: [Select ALL that apply, comma-separated: 1/2/3/4/5/N]

(1. Emotional appeal (uses fear, anger, or pride to bypass rational thinking) / 2. Name-calling or smear (attacks or demeans a person/group with labels) / 3. False context or decontextualization (real image or quote used in a misleading context) / 4. Appeal to authority (implies legitimacy through symbols, uniforms, or institutions) / 5. Simplification (complex issue reduced to a single villain or slogan) / N. Not applicable)

Select N if field 4 is No. If multiple techniques apply, list all (e.g. 1,3).

6. Political preference: [Left/Right/Neutral]

C.6 S6 – Few-shot

You will be shown 4 examples of images with their associated social media context and the correct analysis. Study the pattern, then analyze the final image in the exact same format.

Example 1: [image: 23]

Context: “Vasile-al meu!”

1. Image decision: Yes
2. Detected cues: b,c,d
3. Political propaganda/manipulation: Yes
4. Manipulation technique: 2,5
5. Political preference: Right

Example 2: [image: 640]

Context: “Am convocat astăzi ședința CSAT pentru analiza evoluției situației din Orientul Mijlociu și impactul economic al războiului, în special creșterea prețului petrolului pe piețele internaționale. Una dintre preocupările noastre importante este situația românilor din zonă. Până la acest moment, au revenit în țară aproximativ 5.700 de oameni, dintre care 3.700 au fost ajutați într-un fel sau altul de autorități, prin munca oamenilor din MAE, pe care o salut. România,

împreună cu statele europene, încearcă, prin mijloace diplomatice și politice, să oprească acest război, care are consecințe asupra economiei noastre și asupra românilor care trăiesc în Orientul Mijlociu. Am discutat astăzi și despre dislocarea temporară a unor echipamente și forțe militare americane în România - avioane de realimentare, echipamente de monitorizare și de comunicații prin satelit, acestea din urmă în corelare cu scutul de la Deveselu. Toate aceste echipamente sunt defensive și nu sunt înzestrate cu armament propriu-zis! Pentru ca aceste echipamente și forțe militare să ajungă în România, este nevoie de aprobarea Legislativului. În acest sens, am trimis o scrisoare Parlamentului și, în măsura în care va exista un vot favorabil, ele vor fi dislocate în baza acordului de parteneriat între România și SUA. Așadar, vorbim de o colaborare similară cu cea pe care multe alte state NATO o au deja cu SUA în aceste zile. România rămâne o țară sigură și nu există motive de îngrijorare. Mai mult decât atât, aceste echipamente sporesc securitatea României, care astfel devine și mai bine apărată.”

1. Image decision: No
2. Detected cues: N
3. Political propaganda/manipulation: No
4. Manipulation technique: N
5. Political preference: Neutral

Example 3: [image: 208]

Context: “Dacă el eu este România ! Noi trebuie să-l susținem cu orice preț !”

1. Image decision: Yes
2. Detected cues: a,c,d,e
3. Political propaganda/manipulation: Yes
4. Manipulation technique: 1,4
5. Political preference: Right

Example 4: [image: 705]

Context: “O administrație tânără reprezintă viitorul. Tinerii nu duc în spate păcatele corupției care a măcinat această țară. Vom Reface România printr-o nouă generație de politicieni, pentru o nouă generație de români demni și mândri!”

1. Image decision: No

2. Detected cues: N
3. Political propaganda/manipulation: Yes
4. Manipulation technique: 1,5
5. Political preference: Right

Now analyze the following image using the exact same format. Required strict answering format:

1. Image decision: [Yes/No] (Is the image AI-generated/Deepfake/AI-altered?)
 2. Reasoning — detected cues: [Select ALL that apply, comma-separated: a/b/c/d/e/N]
 - a. Anatomical implausibility / b. Scene incoherence / c. Contextual implausibility / d. Digital artefacts / e. Textual errors in image / N. Real image (no cues detected)

Select N only if image decision is No. If multiple cues are present, list all (e.g. a,d,e).
 3. Political propaganda/manipulation: [Yes/No]
 4. Manipulation/Propaganda technique: [Select ALL that apply, comma-separated: 1/2/3/4/5/N]
 - (1. Emotional appeal / 2. Name-calling or smear / 3. False context or decontextualization / 4. Appeal to authority / 5. Simplification / N. Not applicable)

Select N if field 3 is No. If multiple techniques apply, list all (e.g. 1,3).
 5. Political preference: [Left/Right/Neutral]
- [image: query] Context: {context}

D Additional results

The following tables report Accuracy, Precision, Recall, and Macro F1 for all 36 experimental conditions (three models $\times$ two languages $\times$ six prompting strategies). All values are expressed as percentages. The previous appendix section **C** included the full prompts for all six strategies, in English.

S1 (at **C.1**) represents a simple prompt replicating the prompt an usual non-expert user would use.

S2 (at **C.2**) builds onto S1 and adds the text accompanying the image on Social Media to provide more context.

S3 (at **C.3**) represents the Chain-of-Thought prompt.

S4 (at **C.4**) represents a Persona-building prompt.

S5 (at **C.5**) represents the combination of S3 and S4.

S6 (at **C.6**) represents a few-shot condition prompt, where 4 images, 2 real and 2 AI-generated are provided as labelled examples during each prompt.

Table 4: Accuracy (%) per model, language, and strategy.

Model	Lang	S1	S2	S3	S4	S5	S6
GPT	EN	70.59	69.03	68.61	66.42	70.77	81.08
	RO	56.34	64.82	63.04	65.08	65.44	79.00
Gemini	EN	80.67	70.08	68.58	69.40	69.58	85.17
	RO	73.92	68.58	66.42	72.98	68.75	81.92
Qwen	EN	78.34	73.00	69.00	72.33	70.75	78.67
	RO	87.83	75.10	72.00	73.48	70.33	77.67

Table 5: Precision (%) per model, language, and strategy.

Model	Lang	S1	S2	S3	S4	S5	S6
GPT	EN	100.00	99.12	99.07	99.00	99.07	97.70
	RO	100.00	100.00	99.35	99.45	98.44	77.02
Gemini	EN	97.18	98.79	95.88	99.15	97.57	84.82
	RO	98.98	98.69	99.00	97.92	99.13	77.96
Qwen	EN	99.13	99.29	97.50	100.00	98.82	89.14
	RO	96.14	100.00	95.52	98.96	94.53	99.70

Table 6: Recall (%) per model, language, and strategy.

Model	Lang	S1	S2	S3	S4	S5	S6
GPT	EN	41.14	38.29	38.38	33.17	41.25	63.67
	RO	12.69	29.93	26.37	30.33	31.61	82.67
Gemini	EN	63.17	40.67	38.83	38.93	40.17	85.67
	RO	48.33	37.67	33.17	47.00	37.83	89.00
Qwen	EN	57.05	46.33	39.00	44.67	42.00	57.83
	RO	78.83	49.29	46.17	47.50	43.17	55.50

Table 7: Macro F1 (%) per model, language, and strategy.

Model	Lang	S1	S2	S3	S4	S5	S6
GPT	EN	67.79	65.78	65.56	62.24	67.88	80.49
	RO	46.06	59.98	57.31	60.29	61.00	78.97
Gemini	EN	80.06	67.25	65.53	66.23	66.70	85.16
	RO	72.09	65.26	62.24	71.03	65.44	81.83
Qwen	EN	77.30	70.93	65.93	70.04	68.12	77.69
	RO	87.73	73.19	70.00	71.56	67.97	76.52

Table 8: Proportion of image + text associations flagged as containing political propaganda (%), by model, language, and prompting strategy. Values represent the percentage of parsed responses where the model assigned a propaganda verdict of *Yes*. S1 does not include a propaganda detection field and is therefore excluded.

Model	Lang	S2	S3	S4	S5	S6
GPT	EN	82.5	70.0	86.5	70.3	73.5
	RO	77.1	64.3	66.5	64.4	66.8
Gemini	EN	81.5	61.3	85.9	68.0	51.8
	RO	75.2	59.3	71.8	72.7	57.2
Qwen	EN	76.4	60.3	87.1	69.8	90.1
	RO	72.3	56.8	71.1	61.8	69.0

E Additional visualisations

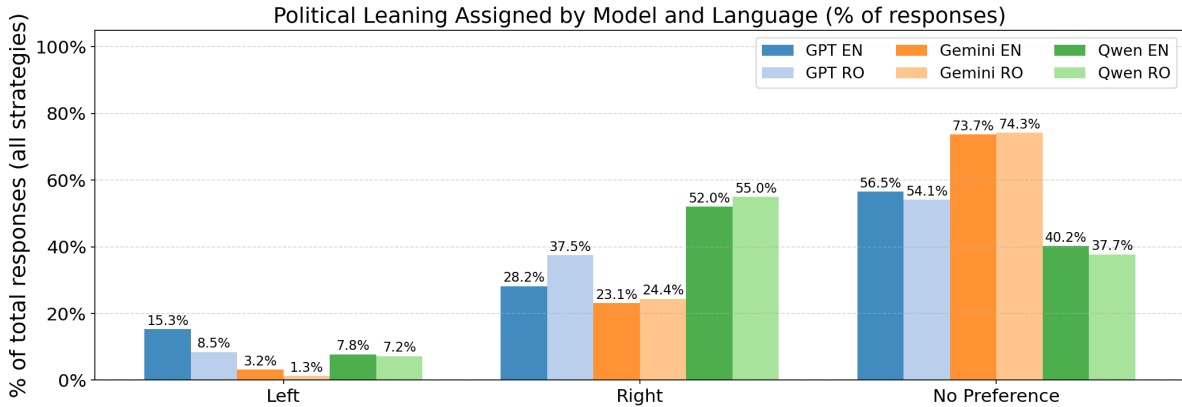


Figure 19: Political preference predictions aggregated by model

Word Clouds: Step-by-step CoT Reasoning by Language

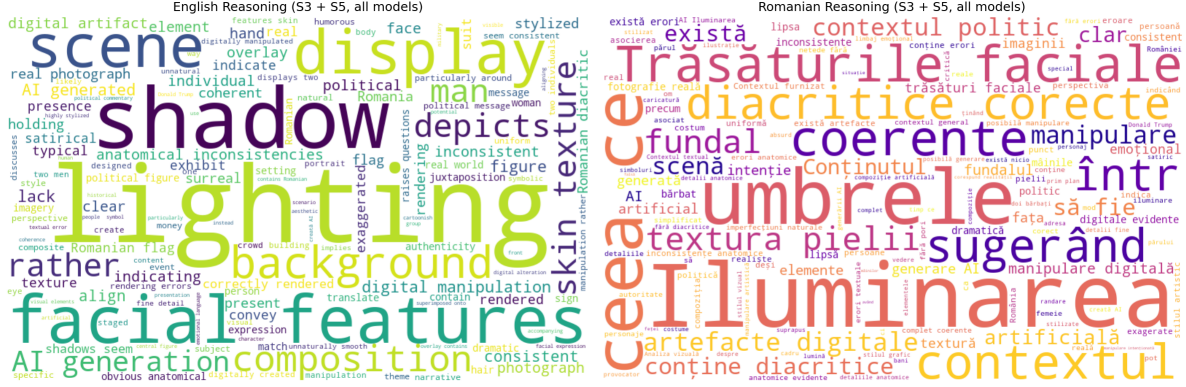


Figure 20: Word clouds for reasoning answer



Figure 21: Example images at varying levels of cross-model disagreement. (a) A fake image that received a perfectly split verdict across all 36 runs. (b) A fake image near the mean disagreement score for the fake class (0.44). (c) The real image with the highest disagreement score in the dataset, where a majority of runs incorrectly classified it as fake.